bitdispatcher manual

A free 911 computer-aided dispatch system in one program, set up entirely in the web interface: the dispatcher screen, the recorded map and aerial photos, road-routed recommendations, move-ups, the mobile terminal and reports built from the ledger.

106 chapters · 6 figures · 22 thousand words · BioDream LLC

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PART 1

Dispatching

The dispatcher’s screen and everything done from it: calls, incidents, units, the map and the command line.

CHAPTER 1

Getting started

bitdispatcher is a computer-aided dispatch (CAD) system for 911 centres: one program with its own web interface. It creates and tracks incidents, recommends units by drive time over the road network, shows units, calls and incidents on a dispatch map with aerial photos, gives field crews a mobile terminal, and records every action in a tamper-evident ledger that reports are built from.

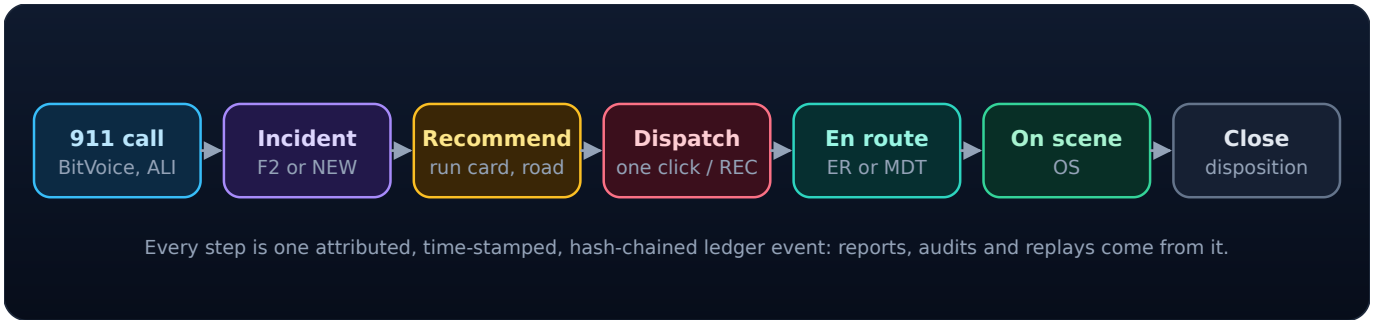


Figure 1. From a 911 call to a closed incident, and what each step writes to the ledger

The first hour

1. **Sign in** as the administrator with the password printed on first start, and choose **Administrator**. See Signing in (chapter 2).
2. **Record the map**: Admin › Map, choose the centre and radius, press Record. About two minutes for 100 miles. Open Map (chapter 94)
3. **Stations and units**: where units are based and what they are. Open Stations (chapter 90) Open Units (chapter 89)

4. **Call types and run cards**: what each kind of incident needs. The defaults are a sensible start. Open Call types (chapter 92)
5. **Users**: dispatchers, supervisors and field crews. Open Users (chapter 99)
6. **Call handling**: connect BitVoice, or point any call-handling system's ALI spill and EIDO at bitdispatcher. Open Integrations (chapter 97)

Admin › Setup checklist follows these steps and the rest of what comes before the first live call, and Admin › Go-live check tests the result. See Self-service setup (chapter 83). Open the checklist (chapter 84)

> bitdispatcher advises; the dispatcher decides. No feature dispatches a unit, closes an incident or changes a priority on its own.

Demo mode

Started with `-demo`, bitdispatcher seeds a fictional agency on the Roanoke, Virginia map (stations at real station positions, made-up units). With `bvsim`, a simulated BitVoice, calls ring and radios report positions; everything simulated is marked SIMULATED.

CHAPTER 2

Signing in: your duty, codes and the lock

Type your username and password. If you hold more than one duty, choose the one you are working: **Call-taker**, **Police**, **Fire** or **EMS dispatcher**, **Dispatcher (all disciplines)**, **Supervisor**, **Administrator** or **Records** (number keys choose). Your workspace opens for it, with your saved preset for that duty (Your own layout (chapter 6)) and your usual agencies (The agency you are working for (chapter 7)), and what you may do follows it: the server enforces it, not only the screen.

Duty	Opens	May
Call-taker	Call entry (chapter 3)	Enter and update incidents; not send units (unless the agency allows it)
Police, fire, EMS dispatcher	The console with that discipline	Send and change their discipline's units (any, if the agency allows it)
Dispatcher (all disciplines)	The console	Everything on the floor
Supervisor	The console, the board (chapter 40)	Also intervene anywhere, reopen, read the ledger, run exercises
Administrator	Administration	Configuration; works the floor only when signed in for a floor duty
Records	Records (chapter 44)	Review views and records requests; changes no incident

A supervisor covering a position keeps supervisory rights. You are never asked which console you are at: an enrolled workstation (chapter 100) gives the position; a browser that is not one keeps a web position of its own (WEB- and four letters).

One-time codes. When the agency asks for them (Settings › Roles and sign-in), the first sign-in shows a QR code: scan it with any authenticator app and type its six digits. After that, type the current code each time; a code works once. An administrator resets it for a lost phone.

The lock. Ctrl+Shift+L, or the agency's minutes without input, locks the console: nothing is shown or changed until you unlock it with your password (and code). Everything stays open underneath, including what you were typing. Signing in and out, locking and unlocking are all in the ledger, with the duty and workstation.

> Before relying on these for criminal justice information, confirm your state's CJIS and access requirements.

CHAPTER 3

Call entry: the call-taker’s workspace

In the web interface: `/taker`

A calm workspace for one call: the call list on the left, nine tabs across the top with only the chosen one shown, and the call (its location, transcript and what was heard) with the **narrative** always on the right.

Several working calls

To the left of call entry, **Working calls** holds each call in progress in its own tab: up to the number the agency allows for the duty (three for a call-taker unless it says otherwise; a trainee gets the agency's number, locked). Each keeps its own caller, location, callback, type, people, vehicles, comments, searches and checks; moving between them changes nothing in the others.

- **New working call** (or **Next call / F2** while one is under way and a tab is free) starts another without clearing the first. A call chosen from the list while one is being written opens in a new tab.
- Each tab shows its number, the type or incident, the place, how long it has been open, **ACTIVE** for the one you are in, **SENT** once submitted, and **!** when it still needs a type or address. Keys and Submit act only on the active tab.
- **Submit** turns only that tab into its incident; the others stay exactly as they were.
- Closing a tab with information in it asks first and asks why; the ledger records each working call opened, moved, submitted or discarded.
- They are kept on the server with you: after a reload, a lock or a lost connection they come back at any console you sign in to. A supervisor can see who holds which and give one left behind to someone else (with a reason), for the agencies they work.

Key	Does
F4 or Ctrl+Enter	Submit the incident; after it, save changes to it
F2	Clear for the next call (asks first if anything is unsubmitted)
F6	Rebid the caller's location
F8	Write in the narrative
F3	CAD search
Alt+1 … Alt+9	Change tab

Call Entry. Type a code or a word for the call type (hundreds are fine; arrows and Enter choose), the priority (the type's, or another), **modifiers** (weapons, in progress, entrapment…: some raise the priority, some add people to notify, officer-safety ones show red), the incident's location (pick a suggestion, right-click the map, or use the caller's ALI position; **Verify** when the caller

confirms it), the caller and callback, and where the caller is when not at the incident. The call's ANI/ALI is beside it.

Before a second incident. F4 first looks for open incidents at the same address, nearby or from the same callback. If there are any you see them: **Add this call to it** puts the call (and what you wrote) on that incident; F4 again makes a new one.

After submission the workspace stays on the incident. Dispatch has it at once; keep typing: comments go straight on it, and F4 saves changes to the type, priority, location or caller. A newer caller location (a rebid, a location provider) shows at the top: **Use it** changes the incident's location (the old one stays in the timeline, dispatchers and units are told); **Keep** leaves it.

The other tabs. Persons and **Vehicles** (several of each, with their role; what the caller said is offered to fill them), **Previous** (earlier incidents linked by address, phone, names, plates and VINs, confirmed or possible), **Hazards** (premises at and near the place, officer-safety people and modifiers, BOLO matches: the critical ones also show as one red line on every tab), **CAD Search**, **Comments** (every note in order, and logging mutual aid and resource requests, transfers, referrals, outbound calls and instructions given), **History** (units and their times, radio, the response recommended) and **Resources** (the agency's approved guidance (chapter 4)).

Notes written before submission keep their own time. Nothing is lost on a lock or a reload: the entry is kept in the browser tab until it is submitted or cleared.

CHAPTER 4

Resources: approved guidance with the call

As the call type, modifiers and what is said change, the **Resources** tab finds the passages of the agency's approved policies, call-taking guides, response guides and checklists that apply, and lights up when there is something new. Each passage shows its document, section, version and effective date, whether it is **policy (mandatory)**, a guide, a checklist or suggested prompts, and why it matched; **The whole document** opens the full approved text at that section. Outdated passages and two policies that both apply are marked. When nothing applies it says so.

Only approved text is shown, word for word; nothing is generated. Opening the tab records which passages, at which versions, were shown for the incident, so a reviewer later sees what was available at the time. Documents are kept in Approved guidance (chapter 104).

CHAPTER 5

The dispatcher screen

In the web interface: `/`

Everything a dispatcher needs is on one screen, and every part of it shows something useful.



Figure 2. The dispatcher screen: calls and incidents on the left, the map in the middle, the selected incident on the right

Area	What it shows
Status bar	Calls waiting, pending and active incidents, units available by discipline, the clock, and the health of every interface (BitVoice, radio, ALI, map, ledger)
Queue (left)	911 calls, ringing first, with callback, location, uncertainty and age; incidents by priority with their timers
Map (centre)	Units by status, incidents, caller locations with uncertainty, road routes of recommended and assigned units with arrival times
Unit board (bottom)	Every unit by discipline: status, time in status, incident, AVL age; overdue timers in amber. **Drag a unit onto an incident** in the queue to send it; **right-click** a unit for its statuses, sending it to the selected incident, and moving it up
Right panel	The selected incident (recommendation, units, timeline, caller), or when nothing is selected the situation: what waits for units, overdue timers, move-ups (chapter 21), coverage by station, recent activity
Command line (bottom)	Type anywhere to start a command (see The command line (chapter 35))

Select an incident by clicking it, with `[` and `]`, or from the palette (`Ctrl K`). `Esc` clears the selection.

CHAPTER 6

Your own layout

The dispatcher screen shows only the sections you want, where you want them. Someone new can start with two and add more when ready.

Your saved preset for each duty

Your layout is a **preset** kept with your account, one for each duty you work (call-taker, police, fire and EMS dispatcher, supervisor…). Sign in for a duty at any console and your preset for it opens by itself: nothing to rebuild at the start of a shift.

- It is **locked** in normal work: nothing moves, resizes or disappears by accident.
- To change it: the **lock** button (*Layout locked: edit it*), arrange, then **Save & lock**. Choosing a ready-made layout also starts editing. **Cancel** puts the saved one back. Each

save is in the ledger with who saved it and when.
- A **trainee** works the agency's assigned training preset; the layout controls are not shown. Someone with no preset of their own starts from the agency's preset for the duty.
- An administrator can **save the layout as the agency's preset** for a duty (the building button while editing) and **protect** a person's preset so only administrators change it (Admin › Operations rules).
- The screens spread over several monitors are the workstation's (see Several screens (chapter 10)): each console remembers its own monitors, so a preset placed on a console with fewer or other monitors falls back to what fits there without changing what you saved.

Open **Layout** at the top left of the status bar.

- **Ready-made layouts**: three for the duty you signed in as, drawn as small pictures. One click shows one. Every duty has **Getting started**: just the calls and the incident being worked.
- **Sections**: calls and incidents, map, incident, unit board, fire houses, police board and the command line. Click one to add it or take it away, or drag it onto the screen.

While the layout is **unlocked** (the lock button beside Layout, amber when unlocked), each section has a title bar:

To	Do
Put a section beside, above or below another	Drag it (from Layout, or by its title bar) to that edge of the other; the highlight shows where it will go
Make two trade places	Drop one on the middle of the other
Resize	Drag the line between two sections
Take one away	Its **×**
Start again from nothing	**Clear the screen** in Layout

Sections snap into place: nothing overlaps and nothing is left empty. **Lock** the layout when it is right: nothing can be moved, resized or taken away until it is unlocked again. Everyone starts locked on their duty's first ready-made layout.

Each person's layout is kept per duty, in this browser. The status bar and the emergency banners are always shown. The layout lock is not the screen lock (Ctrl Shift L), which hides the screen until you sign in again.

CHAPTER 7

The agency you are working for

Where the centre dispatches for several agencies (Admin › Agencies), the status bar shows the agency you are **working** for: its rules apply to what you do (its incident numbers, call types, timers and response plans). Click it to see and change:

- **Working**: one agency, always shown.
- **Covered**: the agencies you dispatch for at this position.
- **Watched**: agencies shown for awareness only.

Your usual coverage loads by itself at sign-in, from your profile (Admin › Agencies › People) and the workstation's (Admin › Workstations). A change is recorded. You may only cover agencies you are authorised to work, and watch those, or those an administrator gave you or the workstation to watch.

What the board shows

Someone limited to some agencies sees only their part of the shared board: those agencies' incidents and units, incidents with no agency yet, units of other agencies sent to their incidents, and 911 calls until they become another agency's incident. The server enforces it: another agency's incident or unit is refused by a direct link or on the command line, and the attempt recorded. Someone who works every agency sees everything, and can choose **The board shows › Only the agencies covered and watched here** to narrow their own board without losing access.

An incident has a **primary** agency (from its call type, or your working agency) and may have **assisting** agencies. Every unit keeps its home agency: a unit of another agency sent to the incident shows it as assisting by itself. **Agencies…** under the units changes them (a new primary agency needs a reason).

CHAPTER 8

The full console

In the web interface: `/console`

#/console always opens the full dispatcher screen, whatever duty you signed in for: a call-taker can look at the whole board there, and come back to call entry (chapter 3) at **#/taker**.

CHAPTER 9

Dispatching: reasons, warnings, plans and acknowledgement

- **Send rec** on a pending incident in the queue dispatches its recommendation in one press.
- **A reason.** Sending units other than the recommendation asks why (quick choices, or your own words); on the command line write it after a slash: `D E5 / closer by radio`. The reason and the response plan followed are kept with the dispatch.
- **Warnings.** When the location is not verified, is uncertain or is missing, the recommendation says so and you confirm before sending.
- **Response plans.** When a plan (chapter 93) applies, its due order leads; a planned unit that cannot go is struck through with why, and the unit going instead says whose place it takes. When the recommendation changes, the change is shown until you mark it seen.
- **Acknowledgement.** A dispatched unit that has not acknowledged in the agency's seconds flashes **NO ACK**; press it (or `ACK E1`) when it does. A crew acknowledges on its terminal.
- **Call entry facts** on the incident: modifiers, people and vehicles, a changed location (in amber for half an hour), where the caller is, BOLO matches, the talkgroup (`TG FIRE-OPS2`) and notifications still to make, each with **done**.
- **Situation**: the shift **passdown** log, active **BOLOs** (issue, cancel; `BOLO text` on the command line; crews see them) and **facilities** with one-press open / diversion / closed.

CHAPTER 10

Screens on several monitors

A position can spread its work over as many monitors as it has. Each monitor shows one **screen**: a window of the same live picture, arranged for one job. All of a position's windows share one sign-in and one selection: choose an incident on the queue screen and it opens on the incident screen and the map.

Open **Screens** with the monitor button on the status bar, from the command palette, or with **Ctrl+Shift+S**.

Screen	For
Incidents and call entry	911 calls and incidents beside the incident being worked; F2 for a new incident
Fire houses	Every station, its apparatus, who is covering and the empty houses
Police status and mark-outs	Every car: what, where, how long, and when its status check is due
Unit status	The unit board, large
Map	The map alone, following the selection
Incident and situation	The selected incident in full, or what needs attention
Supervisor board	Totals at a glance, read-only

- **Add a screen**, choose its **disciplines** (all, fire, EMS, police, other) and **Open** it. **Open all** opens every one.
- **Detect monitors** (Chrome and Edge) shows the monitors as the computer arranges them; give each screen its monitor and it opens there, filling it. In other browsers drag each window where you want it once: it opens there next time.
- The layout is kept on the workstation for the position, so the next shift at the desk gets the same screens. With **Offer to open these screens at sign-in**, the full console offers them in one click after signing in.
- The browser asks to allow pop-ups (and, in Chrome and Edge, window management) the first time.

Disciplines

A screen limited to some disciplines shows their units, and the incidents that need them: an incident's own call type, the units sent to it, and the units its run card asks for (a crash with injuries is on the fire screen and on the police screen). The status bar shows the screen's disciplines as a coloured chip. Every screen still shows every **emergency** and every 911 call not yet on an incident.

CHAPTER 11

Incidents and call entry screen

In the web interface: `/screen/calls`

The queue of 911 calls and incidents, with the incident being worked beside it at full height, the new-incident form (**F2**) and the command line. It is the call taker's and dispatcher's working screen when the map and units are on other monitors. Everything works as on the full console (see The dispatcher screen).

CHAPTER 12

Fire houses screen

In the web interface: `/screen/stations`

Every station as a card, with every unit that lives there or is covering there.

- **The bays**: one door per unit. Shut and green, it is in quarters and available; open, the stripe is the colour of its status.
- **The state**: IN QUARTERS (all home), PARTIAL, COVERED (a move-up is in) or EMPTY (no unit available in the house, with how long).
- **Tiles**: each unit's status, its time in it, and what it is doing (its incident, or where it is covering). Click, drag onto an incident or right-click as on the unit board.
- **Gaps**: a kind of unit the house should keep in quarters but does not, with the best **Move up** in one click (see Move-ups).
- Across the top: houses with a unit in, empty houses, units out, and each kind of unit available out of its total, with every status in a bar.

CHAPTER 13

Police status and mark-outs

In the web interface: `/screen/police`

Every police car in one table, with the map beside it: what it is doing, where, for how long, and when its **status check** is due.

- **Doing**: its incident (click to open it), or its **mark-out**: what it is doing on its own, such as a meal, court, fuel, a report or a follow-up.
- **Where**: the place given when it was marked out, its incident's address, or the nearest address to its AVL.
- **Status check**: a ring fills as the unit's time runs; amber near the end, red and flashing when the check is due, with how long past. A check is due after the status timer (Admin › Settings), or the mark-out's own time.

- **OK** records the status check (the unit is all right) and starts its timer again. **MO** marks it out; **AV** makes it available.
- The tiles across the top count available, on calls, marked out (by kind), out of service and checks due; click one to show only those.

From the command line

- `MO 2A12 MEAL 315 Church Ave`: mark out, with where. Without a place the unit's AVL position is kept. Add a note after a slash: `MO 2A15 COURT 315 Church Ave / Courtroom 2`. `MO` alone lists the agency's kinds.
- `SC 2A12`: status check, all right. Several units at once: `SC 2A12 2A13`.

Mark-outs are also on every unit's right-click menu and in the unit details, and a crew marks itself out from the mobile terminal (**F7**). A mark-out that can be pulled (a meal, a report) is offered by the recommendation when no available unit can fill a line: it is never sent without a dispatcher choosing it. The kinds, their times and which can be pulled are set in Admin › Settings, with the **longest** each may last: a unit out past it shows **OVER** on the unit board and the supervisor board, and the supervisors get a high-priority message once.

CHAPTER 14

Unit status screen

In the web interface: `/screen/units`

The unit board on its own, with larger tiles that also say what each unit is doing. Click, drag and right-click work as on the unit board of the full console.

CHAPTER 15

Map screen

In the web interface: `/screen/map`

The dispatch map alone. It follows the incident or call selected on any of the position's screens, and draws marked-out units where they were marked out when their AVL is not fresh.

CHAPTER 16

Incident and situation screen

In the web interface: `/screen/incident`

The selected incident at full height (its calls, units, timeline, premises, recommendation and report), or, with nothing selected, the situation: what is waiting, status timers past due, move-ups and activity.

CHAPTER 17

911 calls

Calls arrive from BitVoice's event stream, from EIDO pushed by any call-handling system, and from the ALI spill. The same call from several sources is merged into one line.

- **Ringing** calls show before they are answered, so the dispatcher sees what is coming.
- **Location**: the caller's position and uncertainty circle, the address when there is one, and **STALE** when the location is older than the setting in Admin › Settings. Rebid asks for a new one (BitVoice).
- **+ Incident** on a call creates an incident from it, with the location and callback filled in. Calls already attached show their incident.
- Text to the caller, a video link and live video are on the call when BitVoice offers them.
- Its transcript (chapter 46) and what was heard, translation (chapter 47), the device location and floor (chapter 48), crash data (chapter 53), callers grouped (chapter 50) and non-emergency (chapter 51) calls are described in their own chapters.

CHAPTER 18

Incidents

New incident: `F2`, the New button, + Incident on a call, right-click on the map (the address is filled in), a search result's New incident, or `NEW SF 2001 Crystal Spring Ave` on the command line. An address the recorded map knows exactly is placed on the map at once, so units are ranked by road straight away; anything else stays text to be placed.

The incident panel shows the recommendation, the units on it with their times, the caller and the timeline. Notes (`N text`) and priority notes (`PN text`) go on the timeline; priority notes are shown to every dispatcher and unit.

Close with a disposition (`CL Handled`); units still on it become available. Supervisors can reopen (`RO #26-000123 reason`).

Correct the record (supervisors, on a closed incident) changes the call type, priority, disposition, address or a unit's times without reopening it: say why, and the timeline keeps what each was. Reports already sent (NERIS, NIBRS) are sent again by hand.

Replay (supervisors, on a closed incident) plays it back from the ledger: the units move on the map along their recorded positions (about every 30 seconds) while the timeline fills in, at 10× to 120×, with a slider to any moment.

CHAPTER 19

Unit recommendations

For the selected incident bitdispatcher builds a recommendation from its call type's run card: for each line ("3 × engine", "1 × als"), the available units with that capability, closest first. Units already on the incident fill lines first, and scarce capabilities are filled before common ones so a multi-role unit is not spent where another would do.

- **Drive times** come from the recorded road network: posted or typical speeds, one-way streets obeyed, from the unit's fresh AVL position or its station. The reason says so: "1.7 mi by road, 4 min via Franklin Road".
- **Closed roads and traffic** count at once: a drive goes around a road closure (chapter 24), and a road live traffic reports slow takes longer. The source line under the recommendation says when they changed it ("around 1 road closure, with live traffic").
- Outside the recorded map, or with no road between, the time is a straight-line estimate and is marked `~` .
- A unit covering another station is ranked from where it is posted.
- Notes warn when no unit has a capability (consider mutual aid) or when sending the units would leave a station empty.

The recommended units are picked; **Dispatch** (`Ctrl Enter` , or `REC`) sends them. Click units to change the pick, **Use recommendation** to go back to it, or type `D E1 M2` . The ledger records what was recommended and what was sent.

CHAPTER 20

Advisory suggestions

Beside the incident and on the Situation panel, the advisory panel may suggest something: two incidents that may be the same, a call that may be about an open incident, words that do not fit the call type ("smoke" on a fire alarm, "not breathing" on a fall), a repeat address, a priority 1 still without a unit, a unit on scene not heard from.

- **Why**: open a suggestion to see the evidence it relied on (with its source: unit, caller, dispatcher, machine), what it assumed absent, and the rule in plain words.
- **Cannot determine** is an answer: when sources of equal weight disagree ("smoke" and "no smoke" from two callers), it says so and you decide.
- **Accept** runs the suggested command as you (change the type, attach the call, dispatch the recommendation), or just notes it. **Reject** with a one-click reason; it is not shown again. Both are recorded.
- It advises only. It never merges, retypes, reprioritises or dispatches by itself, and core dispatch works the same when it is off or down.

When an upgrade supplies a new rule (such as *Silent or abandoned 911 call*), Admin › Advisory panel offers it above the rules: **Add it**, then **Save version**. A saved codebook never changes by itself.

CHAPTER 21

Move-up and cover

When a station has none of a kind of unit it normally houses available (its engine is out on a call), the Situation panel shows a **Move-up** box with the move that fills the gap: the nearest unit of that kind by road, from a station that can spare one.

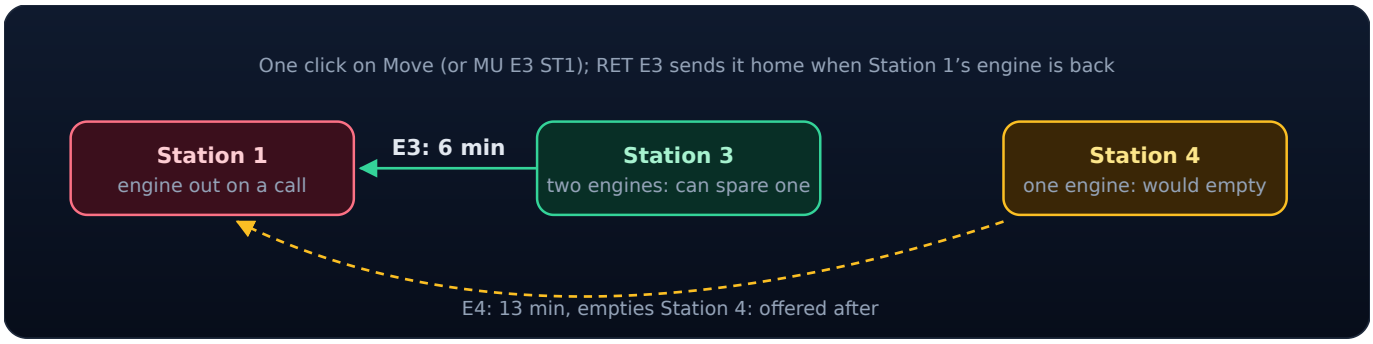


Figure 3. A move-up: Station 1 has no engine, so E4 covers it from Station 4

- **Move E3 from ST3** posts the unit at the empty station in one click (`MU E3 ST1`). It stays available and is recommended from there.
- A move that would leave its own station empty is offered only after moves that empty nothing, and says "empties ST4".
- When the station's own unit is back, a **Home** button sends the cover home (`RET E3`).
- Which kinds are kept covered is set per unit type in Admin › Settings (front-line engines, ladders, medics, ambulances and patrol by default).

CHAPTER 22

Unit statuses

Code	Status	Meaning
AV	Available	In quarters or in service, can be sent
DP	Dispatched	Sent, not yet acknowledged
ER	En route	On the way
OS	On scene	Arrived
TR	Transporting	Taking a patient to hospital
AD	At destination	At the hospital
OOS	Out of service	Not available

Each status has a timer (Admin › Settings); a unit past its timer turns amber on the board and is listed under Status timers past due. Statuses are given on the command line (`ER E1` , `AV E1 M2`), from the unit's tile, or by the crew on the mobile terminal (chapter 37). AVL positions older than the stale setting are drawn dashed and not used to rank units.

CHAPTER 23

The dispatch map

The map is drawn from bitdispatcher's own recorded map (chapter 94): roads by class with route shields and one-way arrows, county and town lines, house numbers and hydrants at street level, fire, police, EMS and hospitals in their own colours.

- **Search** (top right): an address (`2001 Crystal Spring Ave`), an intersection (`Orange & Williamson` , `and` , `/` or `at`), a street, or a place. Arrow keys and Enter; New incident from a result. A house number not on file offers the nearest numbers on the same street.
- **Right-click** anywhere: what is here (the nearest address or road), New incident here, **Close a road from here** and **Close the roads around here** (Road closures (chapter 24)), and Views of this place (chapter 27) when a provider is on.
- **Closed roads** are drawn in red dashes, and **traffic** slowdowns in yellow, orange and red under them; click a closed road for why, and Lift.
- **Routes**: for the selected incident, the road route of each recommended, chosen and assigned unit, with its arrival time on the unit.
- **Day / Night** switches the palette; **Photos** (`F7`) puts aerial photos under the streets; **3D** stands the buildings up (OpenStreetMap's storeys where mapped) and tilts the map, and right-drag turns it; **Fit all** shows every unit and incident; OOS and Stations show or hide those.
- The dashed cyan line is the edge of the recorded map: beyond it the online map is shown, muted, and drive times are estimates.

CHAPTER 24

Road closures and traffic

A closed road changes drive times and unit recommendations at once: the router goes around it, with no rebuild of the map.

- **Close a stretch**: right-click the map on the road, **Close a road from here**, then click along the closed stretch and **Done**. Say why, until when (empty: until someone lifts it), the detour, and whether to note it on the selected incident.
- **Close an area**: right-click, **Close the roads around here**, and give the radius: every road with any part in the circle closes (a flood, a fire, a scene).
- The closed roads are drawn in red dashes. Click one for why and since when, and **Lift** to open it again. A closure with an end time lifts itself.
- A unit can still be sent to a place on a closed road: where a drive starts and ends is never refused, only the roads in between.
- Crews see the roads closed now, with the detours, on the mobile terminal (chapter 37).
- **Live traffic**, when the agency has a feed (Road closures & traffic (chapter 75)), slows the roads it reports: a road's drive time is its recorded time times how much slower it is, up to ten times. The source line of a recommendation says when traffic and closures changed it.
- Every closure, change and lift is an attributed event in the ledger.

CHAPTER 25

Cameras near an incident

When the agency has switched cameras on (Cameras (chapter 76)), an incident with a map position lists **Cameras near**: the agency's cameras within its radius, those looking toward the place first, with how far and whether they look this way.

- Press one: it asks **why** (the incident, or a reason), then opens it. A camera with a still picture shows what it sees now, refreshed every 5 seconds; one with a live view opens it in the camera system, which signs you in itself.
- Every opening is recorded with who, when, why and the incident, and goes on the incident's timeline. Supervisors see the record in Admin › Cameras (chapter 76).
- bitdispatcher keeps no video and records nothing from the camera; the still picture is fetched for the person who opened the camera, and only for half an hour.

CHAPTER 26

Aerial photos

Dispatchers find places by rooftops ("the white house behind the church"). With **Photos** on (F7 or the button on the map), aerial photos are the base and the streets, names and house numbers are drawn over them.

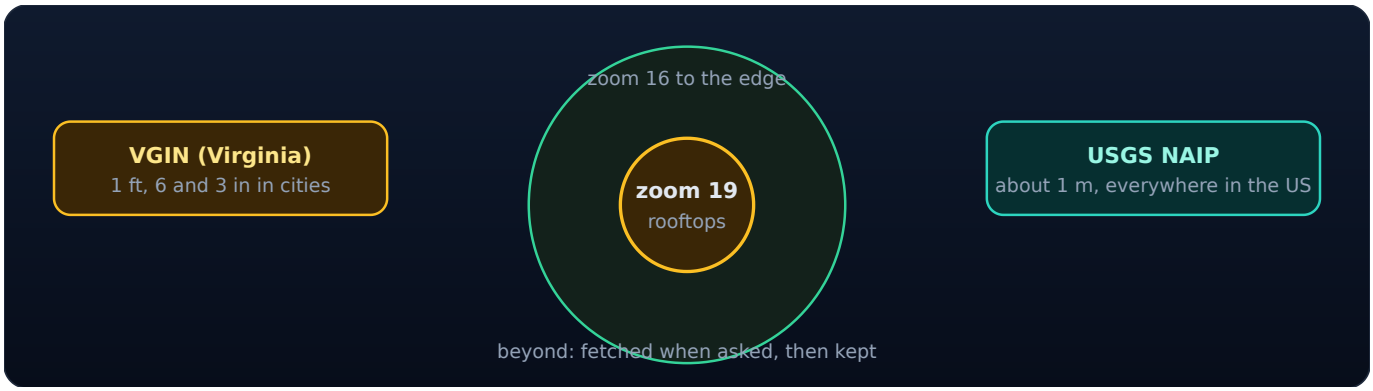


Figure 4. Aerial photos are recorded in two levels of detail

- Photos are recorded in Admin › Map: full detail near the centre, less detail out to the edge. Zooming in past what was recorded enlarges the photo below it, so the map never goes blank.
- The chip beside the button says when the photo at the map centre was taken, where the source publishes it ("2023, 12 inch ortho"): an old photo is never taken for today.

- Where nothing is recorded the server fetches the photo when asked and keeps it, unless that is switched off.

CHAPTER 27

Views: the place from every side

Aerial photos show the roof. **Views** show the front, the back and the sides: where the back door is, the fence, the stairs, the side the hydrant is on.

- **Views** on the incident panel (beside the location), in the call-taker's **Hazards** tab, on the crews' mobile terminal (chapter 37), and from a right-click on the map.
- **N E S W**: the four oblique aerial photos, looking north, east, south and west, from the agency's own EagleView licence. Looking north shows the south side of a building. The arrow keys turn the compass.
- **Street** photos that look at the place, nearest first and at most two per direction, from Panoramax or Mapillary (open photos, CC-BY-SA, coverage depends on who has photographed the street).
- **Pre-plan** photos: the agency's own pictures of a premise (the Knox box, the gas shut-off, the rear entrance), offline and on this server. Crews add them from the terminal with the device's camera, dispatchers and administrators from here or Premises (chapter 102); supervisors remove them.
- Every picture says where it came from and when it was taken, so an old picture is never taken for today. The expand button shows it full size.
- **Google Street View**, when offered, opens Google in a new tab; nothing from Google is shown inside bitdispatcher.

> Looking a place up tells the provider where the incident is. Each provider is off until an administrator turns it on in Integrations (chapter 97), and the pictures pass through this server, so keys never reach a browser.

CHAPTER 28

Messages to and from the crews

Messages go from the floor to the crews' terminals, and from crews to dispatch or to another unit, without the radio.

- **Write** in Messages (Situation panel): to one unit or several (`E1 M2`), a discipline (`ALL-FIRE` , `ALL-EMS` , `ALL-POLICE`) or everyone (`ALL`). Leave To empty to write to the units on the selected incident. **!** makes it urgent.
- **Command line**: `MSG E1 M2 / Stage at the north lot` , `MSG ALL-EMS ! Carilion on diversion` , `MSG #26-000123 / Caller says the back door is open` .
- **From the crews**: messages to dispatch are bold until a dispatcher clicks them, and announced as they arrive. A message on an incident is also written on its timeline.
- **Read**: the terminal marks a message read when it is shown (an urgent one when the crew presses Got it); the floor sees *read by E1*, or *not read yet* after two minutes.
- Every message and every read is a ledger event. Crews write to dispatch or to units, never to everyone.

For messages with a subject, priority, attachments, expiry and versions, and for BOLOs and officer-safety notices, use Messages and BOLOs (chapter 29).

CHAPTER 29

Messages and BOLOs: the inbox

An inbox like email inside CAD, for **messages**, **BOLOs**, **officer-safety** and **administrative** notices (the envelope in the status bar; the agency can add its own kinds).

Writing one

Compose: the kind, the **priority**, a subject and the text (the essential warning goes in the text, never only in an attachment), **To** and **CC**, the owning agency, the incident and related incidents, and for a BOLO the plate, vehicle, person and description. Addresses can be units, `ALL` , `ALL-FIRE` , `ALL-EMS` , `ALL-POLICE` , `DISPATCH` , an agency (`@agency:RPD` : its CAD positions and its units' terminals), a role, a duty, a named person or a console. A BOLO with no one named goes to every position and terminal of the owning agency. **Attachments** (pictures, PDFs and the types the agency allows) are checked by their content, scanned where the agency runs a virus scanner, and kept with their hash. **Effective** and **expires** dates; *no expiration* where the agency allows it. **Save draft** keeps it under Drafts.

Priorities

Priority	Shows and sounds
Urgent	An interruptive banner and a repeating alarm until acknowledged; supervisors are told when no one acknowledges it in time
High	A banner and a distinct tone; acknowledgement may be required, repeated and escalated
Medium	Highlighted, one tone; unread until opened
Low	An ordinary unread mark, a soft tick or silence

The agency sets each level's tone, repeat, acknowledgement and escalation (Admin › Operations rules, where **Play** lets you hear each one). The tones differ by rhythm and loudness, not only pitch: the alarm is five quick pulses, the alert three, the chime two soft notes, the tick one quiet pulse, each with harmonics so it carries over a room and through a headset. Several arriving together sound once, with the highest one's tone. Every priority is shown in words and with an icon, never by colour alone; only levels the agency allows can be silenced.

Browsers keep a page silent until someone has used it. After a reload, the first key or click turns sound on; until then an urgent banner says **the browser is keeping this console silent** with a button to turn the sounds on.

Folders, receipts and versions

Inbox, **Drafts**, **Sent**, **Active BOLOs** and **Archived** (expired, cancelled and replaced ones, labelled as such). Delivery, opening and acknowledgement are recorded for the person and the duty they were working, not just the screen. **Update** sends a new version and **Cancel** a cancellation; earlier versions are kept, shown in the history, and stay on the incident's timeline with the attachments (not uploaded twice). A person sees a communication only when it is for them, or as a supervisor or records staff of its agency. Everything is in the ledger.

CHAPTER 30

Search: incidents, BOLOs, people, vehicles and units

One search (the magnifier in the status bar, or Recall in the classic keys) across incidents (number, type, address, narrative and comments), messages and BOLOs (every version), people

(names in any order, aliases, birth dates, phones, addresses, identifiers, descriptions), vehicles (plate, partial plate with `?` for a character you do not know, VIN, make, model, colour), attachments (file names, captions, and the words inside them: plain-text files, the text of PDFs, and the words in pictures where the server reads them) and units (call sign, radio ID, who is aboard).

- Results are grouped, with filters for the kinds, agency, dates and status.
- Each says **why** it matched (same plate, partial VIN, shared phone, similar description, *in the file's text* with the words around it) and how strongly: **exact**, **partial** or **similar** (similar spellings when asked).
- **Possible matches** are marked as such: a search never links a person, vehicle or incident to anything.
- Expired, updated and cancelled BOLOs say so in capitals; an old warning never looks current.
- You search the agencies you cover and watch; **beyond my agencies** only where you may work them, and every result names its agency. Juvenile names and medical notes are withheld from people without the right, and nothing restricted is counted or shown.
- Every search, and every record opened from it, is recorded in the ledger.

Unit recall (from a unit in the results, or a unit's History): one unit over a period, in one timeline: its statuses and mark-outs, the incidents with their type and address, who staffed it and in what position, status checks, messages, and where it was by AVL or radio. Narrow it by responder, status or incident; download it as CSV.

CHAPTER 31

Alarms, plate readers and gunshot detection

Alarm companies, plate readers and gunshot detection send their alerts to bitdispatcher (Admin › Integrations; the contract is docs/ALERTS.md). Each arrives in the 911 calls list as a call, **data only**, with what it is at a glance: the alarm type and zone, VERIFIED when the alarm company verified it, the plate and its hotlist, the number of rounds.

- Select it: the incident panel shows everything the sender gave (premise, account, permit, keyholder, the operator's call-back line, the camera and heading, a link to the reader's picture).
- Make the incident from it, or attach it to one, as from any call; the agency's own alarm and shots-fired types are suggested when they exist.
- An update from the sender changes the same call; a **cancel** from an alarm company goes on the incident as a priority note.

- Alarm companies are told what became of their alarm: accepted, dispatched, on scene, closed.
- Each alarm is checked against the agency's **permit register**: *Alarm permit P-1: valid*, *EXPIRED*, *SUSPENDED*, or *none on the register*. Its address's false alarms in the last year are counted; at the agency's limit the note says **REPEAT FALSE ALARMS** (Alarm companies & permits (chapter 77)).
- **Phoned alarm** (above the 911 calls): when an alarm company's operator calls on the phone, type its alarm on the alarm card (company, operator, event number, alarm, zone, premise, address, permit, keyholder). It goes in the queue like a sent one, with the same checks, and says who typed it.
- An alert nobody acts on leaves the queue after 30 minutes; it stays in the ledger.

> ASAP over Nlets, the national standard for alarms, is not connected yet: it needs the standard's schema and the state's message switch. Until then each company can have its own address and token over HTTPS (Alarm companies & permits (chapter 77)).

CHAPTER 32

The station screen and toning out

In the web interface: `/station`

When units are dispatched, their stations are toned out (where an administrator turned it on in Integrations). In the firehouse, a screen on a read-only account at **#/station** (choose the station once; it is remembered) shows the station's units and their statuses, and on an alert:

- rising **tones**, twice as many for a priority 1, then the call **read aloud** twice: the units, the call type and the street;
- the call in large type: priority, type, address, the units, the latest priority notes;
- **Acknowledge** stops it and puts *ST1 acknowledged the alert* on the incident; **Read it again** repeats it.

Press **Press to turn the sound on** once after the screen opens (browsers play sound only after a press). The inactivity lock leaves a station screen on a read-only account alone.

- `TONE` on the command line tones out the stations of the units on the selected incident again (`TONE E1`, `TONE #26-000123`); each tone-out is on the incident's timeline.
- An alerting controller from a vendor gets the same alert at its address (docs/STATION-ALERTING.md).

CHAPTER 33

Incidents with neighbouring agencies

Incidents go to and come from neighbouring agencies' CADs as EIDOs, the NENA standard (Admin › Integrations › Neighbouring agencies; docs/CAD-TO-CAD.md).

- **Send**: on an open incident, **Send to a neighbouring agency…**, choose the agency and say why (*mutual aid, need a ladder*; *on your side of the line*). It goes with the location, the callback, our units on it and the notes; the timeline says it was sent.
- **Receive**: what a neighbour sends arrives in the 911 calls list as a call from that agency, data only, with their incident number, type and notes. Make the incident from it or attach it as from any call.
- Each send is a press: closing and unit status do not flow back on their own yet.

CHAPTER 34

Files and photos on an incident

An incident keeps files: a crew's photos of the scene, a PDF, an audio or video clip.

- **Add**: on the incident panel, **Add** (with a caption if you like); on the mobile terminal, **Photo or file** opens the camera. Crews add to incidents they are on. Up to 25 MB each: pictures, PDF, plain text, MP3, WAV, MP4, WebM. The server goes by what the file is, not its name.
- Pictures show as thumbnails (click for full size); others open or download. Each file is on the timeline with who added it.
- Each file's SHA-256 is in the ledger: a file changed on disk is refused, not shown.
- A supervisor removes a file with why. It stays in the record, hidden from everyone below supervisor.

CHAPTER 35

The command line

Type a letter anywhere on the dispatcher screen and the command line takes it (or press `/`). `Tab` completes, arrow keys recall earlier commands, results stay on screen. `HELP` lists every verb.

Command	Does
`D E1 M2`	Dispatch units to the selected incident (`D E1 #26-000123` names one)
`REC`	Dispatch the recommendation
`ER E1` · `OS E1` · `TR M1` · `AD M1`	En route, on scene, transporting, at destination
`AV E1 M2`	Available: clears them from their incident
`OOS E5 brakes`	Out of service with a reason
`N text` · `PN text`	Note, priority note
`NEW SF 12 Main St`	New incident (`NEW CHEST P1 …` sets the priority)
`P 2` · `T VF`	Change priority or call type
`CL Handled` · `RO #26-000123 reason`	Close, reopen
`MU E4 ST1` · `RET E4`	Move up to cover a station; send home
`TG <talkgroup>`	Set the selected incident's talkgroup
`BOLO <text>`	Issue a BOLO for the selected incident, or none
`ATTACH <call> #26-000123`	Attach a 911 call to an incident
`ACK`	Acknowledge the emergency in the red bar (the newest)
`ALARM` · `ALARM 3` · `2ND`	Upgrade the selected incident to its next alarm (or a level); `REC` then sends the extra units. See Alarm levels (chapter 65)
`PAR` · `PAR MISSING E3` · `PAR FOUND E3`	Record a roll call of the crews on scene. See Accountability (chapter 69)
`BM UC`	Record a fire-ground benchmark (UC fire under control, PS primary search…)
`TOW` · `TOW heavy North / VA ABC-1234`	Call the next tow company on the rotation. See Tow rotation (chapter 71)
`PAGE <group> / text`	Page a responder group about the selected incident. See Paging (chapter 70)
`DIR water`	Look up the contacts directory. See Directory (chapter 73)

Command	Does
`FI UNIT1.8TH/STEWART..ABC1234`	Field initiate: a traffic stop (or other officer-initiated incident) in one line. With the classic keys, type the line and press `F7`
`US UNIT1.AS`	Unit status in one line, by the agency's status codes. With the classic keys, the line then `F11`

One-line field commands

The identifiers are separated by the agency's delimiter (a full stop unless it chose another), in the agency's order; two delimiters together skip an optional field (Admin › Operations rules › Command line):

- **Field initiate**: unit, location, type, plate, state, comment, plate type, status. `UNIT1.8TH/STEWART..ABC1234` is UNIT1 at 8th and Stewart, the type left to the agency's (a traffic stop), plate ABC1234. It creates the incident with the unit on scene, the plate on the incident, and the stop's own status-check timer, without another screen; it becomes a full incident by changing its type (`T`), keeping everything. In **two-pass** agencies it only marks the unit out on the stop. Where the agency asks, the plate is queried in NCIC at once.
- **Unit status**: unit, status, disposition, location, type, comment. `UNIT1.AS` is UNIT1 arrived on scene of its incident. A unit alone (`UNIT1` then F11) opens the unit status form instead of guessing. A mark-out code works too: `UNIT1.ADMIN..HQ ANNEX...paperwork`.

A retired or unknown code, or a command your duty may not give, is refused by the server and the attempt recorded; nothing changes.

CHAPTER 36

Keyboard

The keys follow the agency's **key profile** for your duty (Admin › Operations rules). **Ctrl /** shows the map for the screen you are on, with a search box. A key does exactly what the same command or button does; the server checks your right either way, and a trainee works the profile as set.

The standard keys:

Key	Does
Any letter	Starts a command on the command line
`/`	Focus the command line
`F2`	New incident
`Ctrl Enter`	Dispatch the units picked for the selected incident
`Ctrl K`	Palette: go to an incident, unit, screen or chapter of this manual
`[` `]`	Previous and next open incident
`Esc`	Clear the selection
`F7`	Aerial photos on or off
`F9`	NCIC inquiry, with the selected incident as the reason
`F1` – `F6` (mobile terminal)	En route, on scene, transporting, at hospital, available, out of service

The classic CAD profile (PremierOne-style)

An agency moving from PremierOne can choose this profile as a starting point and change it:

Key	Does	Shift + key
F1	Home	Help
F2	Queries (NCIC)	Unit-to-unit message
F3	Update	Recall (search)
F4	Audit trail	Comments
F5	Clear	Premise information
F6	Messages and BOLOs	Refresh pending
F7	**Field initiate**: processes the command line	Previous incidents
F8	New incident	Next incident
F9	Dispatch the recommendation	Previous incident
F10	Process the command line	Premise and location
F11	**Unit status**: processes the command line	The 911 call
F12	Submit	Previous plates

PART 2

In the field

The mobile data terminal a crew uses from the vehicle.

CHAPTER 37

Mobile data terminal

In the web interface: `/mdt`

A crew signs in on the vehicle's laptop or tablet with a **field** account (Admin › Users) and the unit's call sign as the position (`E1`). The terminal opens at once.

- **The assignment**: type, priority, address in large type, the road route drawn on the map with its time and the roads it takes, the callback, the other units on it, and the notes from dispatch, newest first. Priority notes stand out.
- **Status buttons**: one tap each, with the usual next status highlighted; `F1` to `F6` on a keyboard.
- **Notes to dispatch** go on the incident's timeline as the unit's.
- **GPS**: the device's position goes to bitdispatcher as AVL every few seconds (allow location when the browser asks).
- **Emergency**: hold the button for two seconds. Dispatch sees the emergency banner and acknowledges it.
- **Photos** on the map, as on the console.
- **Messages**: what dispatch (or another unit) writes to this unit, with a reply line and quick replies (Copy, Delayed, Need PD…). An urgent message fills the top of the screen and sounds until the crew presses Got it; see Messages (chapter 28).
- **Messages and BOLOs**: the communications addressed to this unit or its agency's terminals (messages, BOLOs, officer-safety notices), highest priority first, with the agency's tones, repeats and acknowledgement as on the console. An urgent or high one that needs acknowledging stays at the top until the crew presses **Acknowledge**; opening one shows every version, the plate and description, and the attachments. Opening and acknowledging are recorded against the unit's sign-on; see Messages and BOLOs (chapter 29).

- **Roads closed**: every road closure (chapter 24) in force, with why, the detour and until when, refreshed every 30 seconds; the route drawn goes around them.
- **Take it**: where the agency allows crews to attach themselves (Settings › Roles), an available unit takes a waiting incident from the list in one press; it is recorded as a dispatch by the crew, self-attached.

A crew acts only as its own unit. Dispatchers can open the terminal at #/mdt to see what a unit sees (the position names the unit).

CHAPTER 38

Incident reports (RMS)

In the web interface: `/rms`

Officers and dispatchers write incident reports at #/rms, or with **Report** on an incident (the incident's location and time are filled in). A crew on a mobile terminal writes its own from the car.

- **Sections**: when and where (location type, whether it was cleared), offences, people, property, vehicles and the narrative. Codes are shown in words and picked in one click; the screen asks only for what the offence needs (a weapon for an assault, the method of entry for a burglary, the drug for a seizure, the injury and the relationship to each suspect for a crime against a person).
- **Saved as you type**: a draft saves itself a moment after typing stops.
- **Before it is submitted** lists what is still missing for incident-based reporting: the FBI's NIBRS data elements and edits, with Virginia's additions (property quantity, the licence plate reader indicator, drug types). Submit is available once the list is empty.
- **Approval**: a supervisor other than the author approves it, or returns it with what to fix. Every save and decision is a ledger event.

The state file

Supervisors see **State IBR file** under the list. Choose the month and press **File**: bitdispatcher writes the month's approved reports in Virginia's incident-based reporting format (the FBI's position-based flat file with Virginia's additions, 2025.2 specification): Group A incidents with their offence, property, victim, offender and arrestee segments, Group B arrests, or a zero report when there is nothing. Upload it to the Virginia Crime Repository by the 15th of the next month. Reports that fail the checks are left out and listed. Making the file is a ledger event, with its SHA-256.

Format: FBI national writes the same reports in the FBI's own flat-file layout, without Virginia's additions, for states that take the national file. That file has been checked with SEARCH's NIBRS Pre-Certification Tool validator, which applies the FBI's edits (`scripts/nibrs-validator.sh`); passing it is not FBI or state certification.

> The settings need the agency's nine-character ORI (Admin › Settings). Before the first production upload, the Virginia State Police ask for a test file by email (dart@vsp.virginia.gov).

Draft from the incident (on a report written from an incident) fills the narrative with a first draft: the type, place, times, units, people, vehicles and notes from the record. With the language model on (Admin › Integrations) the model writes it from the same facts and is told to add nothing, writing *[to be completed]* where something is missing. Either way it is a proposal: check every line and finish it in your words.

CHAPTER 39

NCIC inquiries

NCIC (on the reports screen, F9 on the dispatcher screen, or on the mobile terminal) sends an NCIC 2000 inquiry through the state's message switch: wanted person (`QW` : name with date of birth, or name, sex and race), vehicle (`QV` : plate and state, or VIN), gun (`QG`) and article (`QA`) by serial.

- Each inquiry needs a **purpose** code and a **reason** (the incident number): CJIS requires both.
- The ledger keeps who asked, when, the purpose, the reason and whether a record came back. The terms are kept only as a fingerprint and the answer not at all.
- **A hit is a lead**: confirm it with the entering agency before any action.
- Administrators set the agency's ORI and the switch in Admin › Settings: off, the **training simulator** (plates, names and serials starting with TEST return a record; every answer says SIMULATED), or a switch adapter by TCP.

PART 3

Supervising

Reports built from the ledger, and the ledger itself.

CHAPTER 40

The supervisor board

In the web interface: `/board`

The whole picture at a glance, for a supervisor's second monitor or a wall display. It only shows: nothing on it changes anything.

- **Across the top**: 911 calls not yet on an incident (red when one is ringing), incidents waiting for a unit with their priorities and the oldest wait, active incidents, units available out of those in service, units marked out by kind, status checks due, and empty stations.
- **Where the work is**: the map shaded by how much is open in each area (open incidents plus twice those waiting), with each area's open, waiting and available units. Areas are the county's emergency service zones or beats when they are loaded (Admin › County GIS); otherwise each station's area, the part of the map nearest it. An area with incidents waiting and no unit free, or two waiting, is **SWAMPED**.
- **By discipline**: a ring of every status with the number free, and each kind of unit available out of its total.
- **Busiest areas**, worst first, and **new incidents** over the last six hours by priority.
- **People**: each signed-in position's workload and breaks, and the care list after critical calls. See Breaks and workload (chapter 58) and Critical calls and care (chapter 57). A possible line attack shows across the top (TDoS (chapter 63)).

Choose the disciplines across the top, or open it from Screens with them. **Full screen** is beside the clock.

CHAPTER 41

Operations report

In the web interface: `/admin/reports`

Admin › Operations report builds the Daily Operations Report and shift debrief for a period from the ledger alone: nobody retypes what CAD already holds.

- **Periods**: this shift, last shift (from the shift change times in Settings), today, yesterday, last 7 days, or any range.
- **Contents**: 911 calls, incidents, dispatches and closes; incidents by hour, type, priority and disposition; response times (call to dispatch, dispatch to en route, call to on scene, on scene to clear) as median and 90th percentile; priority 1 incidents; incidents open at the end (the turnover); status timers run out, once per episode; emergency activations and how fast they were acknowledged; each person's actions; unit demand by phase and utilisation; system and configuration changes.
- **Drill-down**: click any number to see the ledger events it is made of.
- **Certify**: once the period has ended, a supervisor certifies it with optional context. The certification is a ledger event with the report's hash; if anything changes later, the report says it changed since it was certified.

> Activity counts intentional actions only and is never a score: it shows workload, not performance.

CHAPTER 42

The advisory codebook

In the web interface: `/admin/advisory`

Admin › Advisory panel holds the codebook: the rules the panel uses, in plain words, with their thresholds.

- **Mode**: off; **shadow** (evaluated and kept in the ledger, shown to supervisors only, so the rules can be judged before anyone relies on them); **live** (dispatchers see suggestions). A new system starts in shadow.
- **Source precedence**: drag to order unit, caller, dispatcher and machine evidence, strongest first. A stronger source outweighs a weaker one; equal sources that disagree give "cannot determine". A silent alarm panel never outweighs a person who can see smoke.
- **Rules**: switch each on or off, set its severity, distances, times, words and the call types they already fit, and edit its explanation.

- **Test** replays the recorded ledger for the last day, week, month or quarter with the rules on screen, saved or not, and counts what each would have suggested.
- **Save version** keeps the change with a note; every version is listed and can be restored in one click. Decisions (accepted, rejected and why) are listed: rejecting a suggestion is never treated as an error.

Agency scripts

JavaScript script and **Lua script** add a rule that is the agency's own code, versioned, tested and switched between shadow and live with the rest of the codebook. A script defines one function for its hook:

Hook	Function	Returns
Each open incident	`advise(i)`	nothing, or `{title, detail, severity, action: {label, command}}` (or a list): a suggestion; accepting runs the command on the incident, as the person who accepts
Unit recommendation	`recommend(r)`	`{prefer: [unit ids], notes: [text]}`: a preferred unit among a line's next best takes a place; notes are shown with the recommendation
Incident reports	`check(r)`	a list of `{field, message}`: shown as **Agency checks** beside the state's; they advise and do not stop a submission

An incident script sees the number, type and its name, priority, status, address and position, minutes since it was created, notes, units on it, hazards at its premises, caller name and number of calls. **Try it** runs the script once on the newest real incident (or report) and shows what it returned, what it logged with `log()`, and how long it took; **show the input** shows exactly what it saw.

- A script has no files, network, process, environment or code loading: only its input and `log()`.
- It always has a time limit (50 ms unless set, at most 1 s). It runs after the CAD's state is released, so a slow script never holds dispatch up.
- A script that fails or runs out of time is shown under its rule and in the test; the panel and the recommendation carry on without it and say so.

CHAPTER 43

Ledger and audit

In the web interface: `/admin/ledger`

Every action that changes state, takes attention or creates accountability is an event in an append-only ledger: who (user, role, position, session), when, what, on which incident, unit or call. Each event carries the hash of the one before it, so a change anywhere breaks the chain.

- **Verify** checks the whole chain and reports the first broken link, if any.
- **Search** by type, person or text; events are newest first.
- The current state is rebuilt from the ledger at every start, and reports are built from it.

CHAPTER 44

Records and requests

In the web interface: `/records`

The records area reads a finished incident and answers requests for it, all from the same tamper-evident ledger (chapter 43) the rest of the system writes to.

Reviewing a record

The review screen shows one incident four ways, and opening or exporting a view is itself recorded in the ledger.

View	What it shows
As first entered	The incident as it was first created, before any later change
Summary	The short account
Detailed	Everything on the incident
Full audit	Every ledger event, with a hash-chain check: it confirms the number of events and reports any break

Print or **Download text** keeps a copy; both are recorded. Open any incident by its number.

Records requests

A request names a requester, a contact, what is asked for, the incidents, and whether the **summary** or **detailed** view is released. Each request shows the working days left until it is due (amber when close, rose when overdue) and the working time taken so far.

1. **Assemble** builds the release copy from the chosen view and proposes redactions by the agency's disclosure rules; the original record is untouched.
2. For each incident, accept or reject each proposed redaction, giving the **basis** required for anything withheld, or **mark a passage** by selecting it in the record. A live preview shows the redacted copy.
3. **Save decisions** records every decision, accepted and rejected.
4. **Approve and release** is the moment a person authorises the release: accepted redactions are applied permanently and the copy is checked for leaks before it leaves. A released copy can then be downloaded, and its hash is shown.

> Records work needs the records duty, a supervisor or an administrator.

CHAPTER 45

Disclosure rules

In the web interface: `/admin/disclosure`

Disclosure rules are the agency's legally reviewed rules for records requests (FOIA). Nothing is hardcoded as exempt; the system proposes, people decide, and your counsel reviews every rule before it is enabled.

A rule has an id, name, jurisdiction and record types, and matches by:

- **Category**: a named category of the record.
- **Words**: any of a comma-separated list.
- **Pattern**: a Go regular expression.

A **legal basis** is required, with an explanation, an optional effective window, and an enabled switch. A rule stays disabled until counsel has reviewed it.

- **Add the sample rules (disabled)** loads examples; they are examples, not legal advice.
- **Try it** pastes a record's text and shows it with the proposed passages highlighted — amber where the confidence is "review", rose where it is "high" — and a list of each with its text, rule, basis and confidence.

PART 4

What 911 centres asked for

Transcripts and translation, callers grouped and screened, crash data, drones, care for the people, training, quality review, outages and line attacks.

CHAPTER 46

Transcripts and what was heard

When call handling transcribes calls, each call shows its **transcript** live, the caller in white and the call-taker in blue, and **Heard on the call**: the details pulled out of it, so the call-taker types less.

- **What is pulled out**: the address, weapons, vehicles, plates, the direction someone went, how many people, injuries (not breathing, unconscious, bleeding), hazards (fire, trapped, gas, a dog, an electric car) and the caller's name. Urgent ones are red and come first, and show on the call in the queue.
- **Use**: on an address, places the call's incident there (a map search, exact when the map knows the number; check it on the map). Before there is an incident it copies the address.
- **Where transcripts come from**: BitVoice's live transcript when it sends one, its transcript of a recorded call afterwards, or any speech-to-text engine posting to the transcript hook (Hooks (chapter 64)).
- Rules, not a model, pull out the details: the same words always give the same details, and each shows the words it came from.

> A transcript is what a machine heard. The call-taker's own judgement and questions still decide.

CHAPTER 47

Callers in another language

A caller speaking another language is translated line by line on the transcript, marked with the language (**Spanish · translated**); **Show original** shows what was said.

- **The phrasebook** built into bitdispatcher works offline. For **Spanish** it translates the words of emergencies both ways; when it knows too few of a line's words it says **gist only**: get an interpreter.
- **Twelve more languages**: Arabic, Chinese (Mandarin), French, Haitian Creole, Hindi, Japanese, Korean, Polish, Portuguese, Russian, Tagalog and Vietnamese. The language is recognised from its script or its words. A caller's line becomes a **gist**: the emergency words found, in order, with … for the rest ("fire · my house · … · please hurry"). The call-taker's **ready phrases** are said in each of them; typed English is not, because the phrasebook will not make up a sentence it does not have. These phrasebooks were machine-written and have not been checked by native speakers.
- **Say it in the caller's language**: press a ready phrase (Help is on the way, Is the person breathing?, Get out of the house now…) or, in Spanish or with the language model, type in English; the translation shows as you type. It is kept on the transcript, texted when **also text it** is ticked (BitVoice), and the speaker button says it aloud on this console in the browser's voice for the language.
- **A language model** (optional, Admin › Integrations) translates what the phrasebook cannot, on an OpenAI-compatible chat API: one on your own server, or a service under your agency's agreement. What callers say is sent to it, so it is off until an administrator switches it on.

CHAPTER 48

Where the caller is: device location and floor

Each call says how its position was found: **device location** (the phone's own fix, often a few metres, the FCC's dispatchable location), GPS, cell sector or on file.

- **Floor**: a civic floor from call handling, and the **altitude** of a 3-D fix with its uncertainty, which tells which floor of a tall building.
- **Trail**: a caller who moves leaves a dotted trail on the map of every position received, the selected call's brighter.
- **Rebid** asks for a new position; **STALE** marks an old one (Admin › Settings).
- No good position? **Send location link** (Location links (chapter 49)) lets the caller's phone send its own.

CHAPTER 49

Location links: the caller shares where they are

When a call brings no usable position (a landline forwarded from a mobile, a caller who does not know where they are, a poor cell fix), **Send location link** on the call sends the caller a link: texted through BitVoice when the call can take a text, or shown to read out.

- The caller opens it and presses **Share my location**; the phone asks to allow it. Its own position (GPS, a few metres outdoors) comes to the call, marked *Shared by the caller (phone's location, by link)*, and again as the caller moves, about every few seconds while the page is open.
- The call shows the link's state; **end it** stops it. The caller can press **Stop sharing**. A link lasts 30 minutes (Admin › Settings › Calls: location links) and takes at most 900 positions.
- The link holds a long random secret; only its hash is kept. It opens a small page with no sign-in and nothing else on it.
- The link uses the public address in Paging groups (chapter 70): the caller's phone must be able to reach this server at it, with a certificate the phone trusts.

CHAPTER 50

Many callers, one incident

When several people call about the same thing, the queue groups them under **N callers about one place**: calls near each other (300 m by default) within a short time (10 minutes).

- Make the incident from one call; the group then offers **Attach N to** that incident, one press for the rest.
- With an incident selected, the group offers to attach all of them to it.
- Calls already on different incidents are never grouped together. Change the distance and time in Admin › Settings (chapter 96).

CHAPTER 51

Non-emergency calls

Calls on a **non-emergency line**, or whose caller says something that is seldom an emergency (a noise complaint, parking, a power outage), go to the **Non-emergency** list under the 911 calls, with the reason.

- Anything urgent said on the call (a weapon, not breathing, fire) keeps it an emergency, whatever else was said; so does crash data.
- **It is an emergency** moves it back; a person's decision always stands and is recorded.
- **Send the usual reply** texts the reply set in Settings, such as where to report online.
- Set the lines, the words and the reply in Admin › Settings (chapter 96).
- Reports taken by the non-emergency intake (chapter 52) wait here too, marked *a report taken by the non-emergency intake*, with every answer; they do not time out of the list.

CHAPTER 52

Non-emergency intake

In the web interface: `/admin/intake`

A voice or text bot on the non-emergency line (BitVoice's AI receptionist, or another) can take reports. The bot carries the voice; bitdispatcher decides what it asks.

- **Questions**: asked in order, each with the kind of answer (words, an address, a phone number, yes or no). *needed* asks once more when the answer is empty. A yes/no question marked **yes = hand over** sends the caller to a person on yes, or when the answer is unclear. The keys `where`, `phone` and `name` fill the report's address (placed on the map when the recorded map finds it exactly), callback and caller.
- **Handing over**: the moment a caller says one of the emergency words (by default: emergency, hurt, bleeding, fire, gun, breaking in, right now… and the agency's critical words), or anything the transcript reader marks urgent, the bot is told to say the hand-over line and transfer the caller to a call-taker. A word just after *no*, *not* or *nobody* ("nobody is hurt") does not count.
- **A finished report** is a call in the non-emergency list (chapter 51) with every answer, the conversation and a report number the caller is told. Nothing is dispatched by it; a person makes an incident of it or closes it. A conversation handed over is a call marked emergency, with what was said.
- **The bot's token**: **Make the token** (shown once). The bot calls `POST /api/intake/sessions` to start and `POST /api/intake/sessions/<id>` with each thing the

caller says; each answer says what to say next and whether to hand over (docs/INTAKE.md).
- **Try it** runs the saved script in a chat, as a caller would, and makes no report.

CHAPTER 53

Vehicle crash data

A vehicle's automatic crash notification arrives as a call with **CRASH DATA**: the vehicle, delta-V (the change of speed in the crash), the direction of the impact, rollover, airbags, occupants and belts, and its power (electric, hybrid).

- **Injury risk** is high for delta-V of 40 km/h (25 mph) or more, a rollover, or 20 km/h with no belt, several impacts or a side impact; medium for 20 km/h or any airbag. Suggestions follow: ALS, extrication, air medical, more than one ambulance, and for an electric car a word to the fire crews.
- It is a rule of thumb from field-triage criteria to help pick the response, not a diagnosis. Change the thresholds in Admin › Settings (chapter 96).
- Crash notifications reach bitdispatcher through the crash hook (Hooks (chapter 64)).

CHAPTER 54

Drones

With a drone system connected (Admin › Integrations (chapter 97)), an incident on the map shows **Send a drone**. bitdispatcher sends the drone system the incident, its type, priority, position and summary; the drone system decides which drone flies.

- **Launch rule**: incidents of call types marked **Send a drone** (Call types) ask by themselves, once each.
- The drone reports back to the drone hook: its status, position, battery and live picture. It flies on the map, its status goes on the incident's timeline, and **Live picture** opens its video.

CHAPTER 55

Radio transcripts

When BitVoice or the radio system transcribes transmissions, a unit's words go on its incident's timeline in italics (`E1: on scene, smoke showing`), so the record holds what was said on the air. The words come with BitVoice's radio events, or through the radio-words hook (Hooks (chapter 64)).

CHAPTER 56

Incident summaries

Every open incident has a **Summary**: what and where, what the callers said that matters (the urgent first), the priority notes, the hazards on file at the place, who is on scene and who is responding, and a drone over it.

- It is built from the record by rules, so it says nothing the record does not; it changes as the incident does.
- It shows on the incident panel (Copy puts it on the clipboard) and at the top of the crews' mobile terminal (chapter 37).

CHAPTER 57

Critical calls and care

Some calls stay with the people who take them. An incident is a **critical call** when its call type is marked critical (cardiac arrest, shooting and stabbing to begin with) or when words such as not breathing, CPR, officer down or infant are said on it.

- Everyone who worked it (the call-taker and the dispatchers who entered, sent or noted) goes on the **care list** on the supervisor board (chapter 40), with the reason.
- The supervisor **offers a break** to each person, **arranges a debrief** (with a note of when and where) and marks it **Done**. Each is recorded.
- Once the incident closes, each person sees a quiet line on their own console: that was a hard call, take a moment, and the support line set in Settings. **Take a break** puts them on a break.
- Set the words, the support line and the call types in Admin › Settings (chapter 96) and Call types (chapter 92).

CHAPTER 58

Breaks and workload

The cup on the status bar puts you **on a break** and back; the supervisor board shows it.

The board's **People** panel lists every signed-in position: calls in the last hour, actions, time since the last break, critical calls in four hours, and a state:

- **watch**: two hours without a break (Settings), a critical call, or a very busy hour;
- **relieve**: three hours without a break, or two critical calls.

It names the least busy positions free to relieve someone.

CHAPTER 59

Training

Tick **Training** when signing in. The console opens a board of its own, marked TRAINING in violet: the agency's units, stations, call types, premises and map, on a ledger of its own. Each trainee has their own board, even on a shared account; one left idle for two hours is cleared. Nothing done there reaches the live board. Administration there is staging: an administrator's changes stay on the training board until published (see Exercises, injects and staging (chapter 60)); users, connections and maps are the live board's.

- **Scenarios** play calls, transcripts, crash data and radio traffic into it on the agency's own map: a house fire with three callers, a cardiac arrest in Spanish, an interstate crash with crash data, an armed robbery, and a flood of calls on the line.
- **Score** checks what was done: every caller on an incident, the call type, the units sent, a reply in the caller's language, and each incident's quality review (chapter 61).
- **Reset** empties your board and starts it again; other trainees' boards are untouched.

CHAPTER 60

Exercises, injects and staging

An exercise is one training board for everyone who joins it: tick **Training** at sign-in and type the exercise's name (any name; the first to join makes it). Trainees sign in with their own duties, so a call-taker, dispatchers and a supervisor work it together, and crews can join on the mobile terminal (chapter 37) the same way. Practising alone, leave the name empty.

The instructor (a supervisor or administrator) presses **Inject**: a new caller (where, who, and what they say as the exercise goes), the caller saying more or moving, a unit going out of service, an emergency button, or a unit on the radio. **Timeline** downloads every event for the debrief, and when the exercise ends its timeline is kept (Supervisors: `/api/exercises`).

Staging. An administrator signed in with Training changes call types, units, stations, response plans, guidance or facilities on the training board, tries them there, then opens **Staged changes**: it lists what differs from the live board; choose the parts, say why, and **Publish**. With *Change configuration only through staging* on (Settings), those changes cannot be made on the live board directly.

> The training board is not a backup of the live board. A continuity site is a separate, tested arrangement: see Several sites (chapter 101).

CHAPTER 61

Quality review

In the web interface: `/admin/quality`

Every incident checked against the centre's standards:

Check	Standard (Settings)
Answered in time	15 s, NENA's call-answering standard
Entered in time	answer to incident, 90 s
Dispatched in time	entry to the first unit, 60 s, priority 1 and 2
Location verified, callback, disposition	captured
Questions asked	heard on the transcript: the address, the callback, and by discipline (breathing and awake for EMS; anyone inside for fire; weapons, description and direction for police)

Each incident gets a score over the checks that applied; the page lists them for a period, each call-taker's average and each check's pass rate. Press an incident for its checks. A review is a coaching aid: a missed question may simply not be on the transcript.

CHAPTER 62

Working through an outage

Each console keeps the last board it had. When the server cannot be reached, an amber bar says so and the console shows that board, as it was, with the time. A console reloaded during the outage still opens.

- **Offline log**: new incidents, notes and unit statuses are kept in this browser, in order.
- When the server is back the bar offers **Send them now**: each goes in marked "entered offline at 14:02 by jlee"; anything the server refuses stays, with the reason.
- For a server that never goes away, run bitdispatcher at two sites (Several sites (chapter 101)): each keeps working when the other is cut off.

CHAPTER 63

Line attacks (TDoS)

A telephony denial-of-service fills the lines so real callers cannot get through. bitdispatcher watches the calls and raises **Possible telephony denial-of-service** on every console and the supervisor board when:

- far more calls arrive in a minute than usual (five times the last hour's rate, at least 12, or the number set in Settings);
- one number calls five times or more in ten minutes (shown as ×5 on its calls);
- eight or more silent or very short calls arrive in five minutes.

A supervisor **acknowledges** it; it clears after ten quiet minutes. Keep answering: a real emergency can be among them.

CHAPTER 64

Hooks and connections

Other systems send to bitdispatcher with `Authorization: Bearer` and the hook token set in Admin › Integrations (chapter 97). Without a token the hooks are off.

Hook	Sends
`/api/hooks/transcript`	`{"call": "id", "speaker": "caller", "text": "…", "lang": "es"}`, or `lines`: a call's words
`/api/hooks/acn`	a crash notification: JSON or key=value, common telematics names (delta_v_mph, pdof, rollover, airbag_deployed…)
`/api/hooks/drone`	`{"id", "status", "lat", "lon", "battery", "incident", "videoUrl"}`
`/api/hooks/radio-text`	`{"unit" or "radioId", "channel", "text"}`

The drone system's launch address and token, the launch rule, and the optional translation model are set on the same screen.

PART 5

Fire ground, crews and the police desk

Bigger fires, who is aboard each unit, roll calls, paging volunteers, tows, the schedule, the numbers to call, areas on the map, a unit’s day, patient-care reports, papers to serve and alerts to the public.

CHAPTER 65

Alarm levels: 2nd and 3rd alarms

A working fire outgrows its first run card. Each call type may have **further alarms** (Admin › Call types): the 2nd alarm, the 3rd, each adding units to what the incident needs. The first alarm is the call type's own run card.

Upgrading

- On the incident, **Upgrade to 2nd alarm** says what it adds (`+2 engine, 1 ladder…`); or type `ALARM` (`2ND` , `3RD` , `ALARM 3` for a level).
- The upgrade is a priority line on the timeline every dispatcher and crew sees, with what it adds.
- The recommendation then fills the bigger run card, counting the units already there, and `REC` (or Ctrl Enter) sends the extra units. Paging groups and station alerts follow the dispatch as usual.
- **Bring down** goes back a level, with a reason.

> Setting up: Admin › Call types › a type › Further alarms. Each alarm has a name and run-card lines (capability and count). The structure fire type comes with a 2nd and 3rd alarm to edit.

CHAPTER 66

Personnel and crews

In the web interface: `/admin/personnel`

The **roster** is the agency's people: a number, name, rank, qualifications (paramedic, EMT, driver, officer…), home station, and how to page them (a phone for texts, an email, a push topic). Add them one by one or **Import from CSV** (columns: id, name, rank, qualifications separated by ;, station, phone, email, volunteer).

Who is aboard

On any unit (the unit's panel › Crew › change) or from the crew's own terminal (**Crew**), pick who rides it this shift. A member is on one unit at a time: putting them on another moves them. The incident's timeline records crew changes on a unit working it.

What staffing changes

- **Minimum aboard** (Admin › Settings › Unit types, or per unit in Admin › Units): a unit below it is shown short-staffed, in the recommendation's reasons and on the supervisor's exceptions.
- **What a capability needs** (Admin › Settings › Crews): by default ALS needs a paramedic aboard. A medic whose recorded crew has no paramedic is not recommended for an ALS line (it is still BLS and transport).
- A unit with **no crew recorded** is taken as staffed: an agency that does not keep rosters here sees no change.

> Dispatchers see the roster without phone numbers and emails; supervisors see them.

CHAPTER 67

Staffing: who is on duty and on each unit

In the web interface: `/staffing`

Staffing (the people button in the status bar) shows who is on duty, where, and who actually staffs each unit. A call sign alone is not enough: the crew follows the unit into incidents, recommendations, history and reports.

People on duty

Sign on with the kind of duty (regular, overtime, holdover, a temporary detail and where, a call-back), the shift, until when and availability (available, assigned, unavailable, training, court, meal, station duty, administrative). **Sign off** clears the person from any unit. Search by name, qualification, agency, station, unit and availability.

Crews

Edit crew names the people on a unit and their **positions** (driver, officer, firefighter, medic, partner, team leader, or the agency's). The kind of unit may require positions and a minimum (Admin › Settings › Unit types). A crew may sign itself on from the terminal; the agency decides per kind of unit whether that is allowed, needs a supervisor's approval (**Sign-ons waiting**) or is for dispatch only.

- One person is on one active unit at a time. Putting them on another asks: **move them** (relief, a swap) or a **secondary assignment** with a reason.
- Each change has a reason and an **effective** time, kept beside when it was entered.
- A unit left short (below its minimum, a position empty) is **limited**, or put **out of service**, as the agency says, unless a supervisor records an **override** with a reason.
- A lost connection signs no one off.

On incidents and later

When a unit is assigned, the incident keeps who was aboard and what they were qualified for at that moment; later changes never rewrite it, and changes during the incident (who joined, who left, positions) are on its timeline. A supervisor can **correct** a crew record with a reason: the original stays beside the correction, with the entered and effective times. Unit recall and reports show the people as well as the call sign.

Supervisors also see **saved crews** for a **shift change** in one step (each person and time still recorded), and **coverage**: the minimum qualified people the agency wants on duty, gaps, expiring credentials and each unit's capability (a supervisory view the agency controls).

CHAPTER 68

Qualifications, special teams and credentials

In the web interface: `/admin/qualifications`

Each agency keeps a controlled skills file (Admin › Qualifications & teams; supervisors).

- **Qualifications**: special teams (Swift Water, HazMat, Technical Rescue, Dive, K-9, SWAT, UAS), specialties (evidence technician, fire investigator, crisis negotiator, tactical dispatcher), training roles (CTO, FTO, instructor, evaluator) and credentials (EMT, paramedic…), each with its levels, the agency that owns it, how many months it lasts, when renewal is due, its prerequisites, whether it must be **verified** before it counts, and the documents it needs.
- **A person's credentials**: the level, state (trainee, qualified, instructor, evaluator, restricted, temporarily unavailable, suspended, expired), issuer, dates, restrictions and attached certificates. A supervisor marks one **verified**; an unverified self-entry never counts where verification is required. Every change is kept.
- **Equivalencies**: another agency's qualification accepted as one of ours, with its scope and expiry, for mutual aid; the person's own record is never merged or overwritten.
- **Reminders**: supervisors are told once when a credential's renewal comes due and when it expires.

Live capability

A unit's capability comes from the apparatus and the people validly qualified aboard now: a HazMat unit without its required technicians is not HazMat-capable, and shows **LIMITED** with the reason (what each capability needs of a crew is set in Personnel). The recommendation uses the live capability; the agency decides whether sending a short unit is only warned, needs a supervisor's override, or is blocked. Dispatchers see the capability and the qualification names; credential details are for supervisors and the person.

CHAPTER 69

Fire-ground accountability: roll calls (PAR)

At a working fire the incident commander takes a **roll call** (a PAR) of every crew on scene at set intervals, and dispatch keeps the clock.

- A call type marked **Fire-ground accountability** (Admin › Call types) starts the clock when the first unit arrives on scene; **Start roll calls** starts it on any fire incident.
- The incident shows the time since the clock started and when the next PAR is due; it turns amber when overdue, and the supervisor's board lists it.
- **All accounted for** (or `PAR`) records a roll call of every crew on scene. A crew that did not answer: click it, then **Report … missing** (or `PAR MISSING E3`). That is a red, flashing line on the incident and an urgent exception for the supervisor until **E3 found** (`PAR FOUND E3`).
- **Benchmarks** the commander reports (primary search complete, fire under control, loss stopped…) are one click each, or `BM UC`, with the time since the incident started.

- The clock stops when the incident closes; it cannot be stopped while a crew is missing.

> The interval is 20 minutes unless changed in Admin › Settings or on the call type.

CHAPTER 70

Paging responders

In the web interface: `/admin/paging`

Volunteer and off-duty firefighters and EMTs are called by **page**. A **paging group** says who is paged together and how:

- **Members** from the roster (Personnel (chapter 66)).
- **Gateways**: how pages go out, set up by an administrator in Admin › Integrations › Paging: a text by Twilio, an email, a push to phones by ntfy, a webhook, or IamResponding and Active911 (both take dispatch by email in their documented formats).
- **Automatic**: page the group when one of its units or stations is dispatched, or when an incident of one of its call types gets units. Each group is paged once per incident by its rule.

A page

The page's subject is the call type and address, the text the incident's summary and units. Each person gets a link to answer: **Responding to the station**, **Going straight to the scene**, or **Not available**, with when they expect to arrive. No sign-in: the link is the key, and works only for that person and that page.

On the incident, **Responders** shows how many were paged and responding, each person's answer and minutes away; **Page** pages more groups. `PAGE <group> / text` does it from the command line. A page no one has answered after five minutes is on the supervisor's exceptions; failed deliveries are on the timeline.

> Set the **address responders open** (Admin › Integrations › Paging) to the address phones reach this server at; without it pages go out with no answer link. Tested against stand-ins for Twilio, ntfy, email and a webhook; not certified by IamResponding or Active911.

CHAPTER 71

Tow rotation

In the web interface: `/admin/tows`

Police call wreckers from approved companies **in turn**. Each company tows some classes (light, medium, heavy) in some zones (Admin › Settings lists both; one zone for the whole area by default).

- **Next on rotation** on the incident (or `TOW`, `TOW heavy North`) gives the company whose last turn in that class and zone is the oldest, with its number to call, and records the call.
- Mark what they said: **Accepted**, **Declined** or **No answer** (both lose the turn; ask for the next), then **Arrived** or **Cancel**.
- An **owner's choice** or **officer's choice** of company is recorded but takes no one's turn.
- A suspended company is passed over, and the rotation screen says why each company is or is not offered.

Every request and answer is on the incident's timeline and in the ledger, so the rotation's fairness can be shown.

CHAPTER 72

Scheduled calls and special events

Work known ahead of time (the calendar button in the status bar):

- A **scheduled call** (a planned transport, an escort, a business check) becomes an incident by itself at its time, or a set number of minutes early, and sends its units if they are available (those that are not are named on the incident).
- A **special event** (a fair, a parade, a game) becomes a standby incident for its hours, and is listed as running on every console.
- Either can repeat every day, on weekdays or every week, until a last day.

Scheduled today on the situation panel lists what is coming in the next 24 hours and what is running now. An occurrence missed by more than an hour (the server was down) is not made late; it is skipped and recorded.

CHAPTER 73

Contacts directory and on-call lists

In the web interface: `/admin/directory`

The numbers dispatch calls: utilities, agencies, the medical examiner, interpreters, public works. Each entry has numbers with labels (after hours, cell), an email, notes and tags.

- **On-call lists** say who answers tonight: shifts of directory entries by date and time, with a backup when no shift covers the time.
- An entry or list can be on the **call-out list** of call types: every incident of the type lists **To call** with the person on call now and the number (a tap calls it on a phone), and **Called** logs it on the incident. A call type's notifications (Admin › Call types › Notify) are matched to entries by name, organisation or tag.
- The directory button in the status bar (or `DIR water`) searches by name, organisation, tag or number; the situation panel shows who is on call now.

CHAPTER 74

Areas on the map (geofences)

In the web interface: `/admin/geofences`

Areas drawn on the map:

- An **alert area** tells dispatch when a unit enters or leaves it (a unit leaving the county, entering a hazard zone, arriving at a hospital): a notice on every console, the situation panel's **Area alerts**, and a line on the unit's incident. It can watch only some disciplines or units.
- A **caution area** puts its note on every incident inside it (a school zone, a firing range, a block known for trouble), on the console and the crews' terminals.

Draw a new area: click the corners on the map, then Finish. Areas show on the console map, dashed. Whether a unit is inside is worked out from its positions as they arrive, so an area drawn around a unit already there is not an arrival.

CHAPTER 75

Road closures & traffic

In the web interface: `/admin/roads`

Every road closed now, with why, the roads it takes, since when, until when, the detour and who closed it; and those ended in the last day.

- **Lift** opens a road. **Remove** is for a closure entered by mistake (it stays in the ledger); a closure that was real should be lifted, so the record says it was closed.

- A closure whose line or circle covers no recorded road says so: draw it again along the road.

Live traffic (administrators):

- **Waze for Cities** is Waze's free data-sharing programme for public agencies. Once the agency has joined, Waze gives it the address of its traffic feed (JSON); paste it here. Each jam's speed slows the roads under it; a road Waze reports closed is closed.
- **GeoJSON**: any other source, a FeatureCollection of lines with `speed_kmh`, `factor` (times slower) or `closed: true`, and `name`.
- Fetched every 2 minutes by default (30 seconds to an hour), with a token sent as Bearer when the feed needs one. The status says when it was last fetched, how many slowdowns it had, or why it failed. A feed that stops leaves the recorded drive times.

CHAPTER 76

Cameras

In the web interface: `/admin/cameras`

The cameras dispatchers may open for an incident (Cameras near an incident (chapter 25)): traffic cameras, the agency's buildings, schools and partners that agreed to share.

- **Offer cameras near incidents** is off until an administrator switches it on; the radius (400 m by default) says how far from an incident cameras are offered.
- **Add a camera**: its name, owner, where it is, which way it looks (degrees from north; empty if it turns), how wide and how far it sees, and its **live view** address (opened in a new window in the camera system) and/or its **still picture** address (a JPEG of now, fetched through bitdispatcher). Only https addresses, or http on the agency's own network.
- **Who looked, and why** lists every opening. Agree a policy before switching cameras on: which cameras, for which calls, and who reviews this record.

CHAPTER 77

Alarm companies & permits

In the web interface: `/admin/alarms`

Companies: each central station the agency works with gets its own address (`/api/hooks/alarm/<company>`) and token, shown once when it is made; its alarms arrive with its

name, and what became of them goes to its own answer address. A new token stops the old one. Companies not on the list can still use the shared hook token (Admin › Integrations).

Permits: load the agency's alarm permit register as CSV, with a heading row: number, address, holder, company, expires (YYYY-MM-DD), status (active, suspended, revoked). Loading replaces the register. Each alarm is matched by its permit number, else its street address.

False alarms: an address's incidents in the last year whose disposition is a false alarm are counted (by default, dispositions with "false" in them); at the limit (3 by default) the alarm says **REPEAT FALSE ALARMS**.

Phoned-in alarms are typed on the alarm card (Alarms (chapter 31)).

CHAPTER 78

A unit's status history

History on a unit's panel shows its day from the ledger: a bar of its statuses across the day, every status with when, how long, the incident and who changed it, its status checks, crew changes and move-ups, the incidents it worked, and the share of its known time spent on incidents. Pick another day with the date; **CSV** downloads the same.

Unit recall (Search (chapter 30)) gives any period in one timeline for front-line use, with who staffed the unit, its messages and its locations.

CHAPTER 79

Patient-care reports (NEMSIS)

An EMS crew writes its patient-care report in its ePCR system; the dispatch part (when the call came in, when the unit was notified, en route, on scene, left the scene, at the hospital, back in service, where, and the dispatch reason) comes from bitdispatcher.

- **NEMSIS** on an EMS unit of the incident downloads its response as a NEMSIS 3.5 record, to import into any ePCR that takes NEMSIS files.
- With a patient-care report system set up (Admin › Integrations › Patient-care reports), **Send** posts it there, and **automatic** sends each EMS unit's response when it clears. What the system answered is on the incident.
- The dispatch reason is the call type's NEMSIS code (Admin › Call types), or suggested from its name.

> The record carries dispatch, times and the scene only; the crew completes the patient sections. It follows the NEMSIS 3.5.1 schema for those sections; tested against a stand-in, not a vendor's system.

CHAPTER 80

Warrants and papers to serve

In the web interface: `/papers`

The papers a sheriff's office or police department holds to serve: arrest and bench warrants, protective orders, evictions, summonses and subpoenas (the gavel button in the status bar). Each has the person, where to serve it, the court and number, any **caution** (armed, violent, dog), and every attempt.

- An incident near the address of an active paper (or naming its person) shows **active paper at this address** with the caution to every console and to the police crews' terminals; **Details** (police dispatch, records and supervisors) shows the papers.
- **Record the attempt**: the unit, the result (no answer, not there, moved, refused, served) and a note. Served closes it; **Recalled** and **Returned unserved** need a reason.
- Every look at the papers is recorded in the ledger.

> Confirm every warrant with the issuing agency (and NCIC where it applies) before acting on it.

CHAPTER 81

Alerting the public

Supervisors warn the public from the console: **shelter in place**, **evacuate**, police activity, a hazardous spill, a missing child, 911 being down. Press the **megaphone** in the status bar, or **Alert the public** on an incident.

Writing an alert

- **Real alert**, **Exercise** or **Test**: an exercise or test is marked as one everywhere it goes. Check this first.
- **Kind of alert**: the SAME event code (SPW shelter in place, EVI evacuation, LEW law enforcement, HMW hazardous materials, CAE child abduction, TOE 911 outage…).

Choosing one fills the wording, in English and Spanish, to start from.
- **Headline**, **what is happening** and **what to do**.
- **Phone alert**: what phones show first (Wireless Emergency Alerts, 90 characters) and what text messages and push notifications say; the counter turns red past the limit. A **longer phone alert** (360) is optional.
- **Also in Spanish** adds a Spanish version; people on the lists who chose Spanish get it.
- **Where**: around the incident (a quarter mile to two miles), an area drawn on the map (Admin › Geofences), or **everyone** in the jurisdiction (its county codes).
- **Expires after** one to 24 hours.
- **Send to** the subscriber lists and the CAP systems (sirens, alert software) chosen; **keep it off some channels** asks IPAWS alert software to leave out radio and TV, weather radio, phones or public sites.

Check shows the CAP message it makes and anything wrong with it, and how many people it reaches. **Send** asks once more, with the exact phone text and area. With **two-person approval** on, the alert waits until a second supervisor presses **Approve** (the megaphone shows *to approve*); its author can withdraw it, and the second supervisor can reject it with a reason.

In force

An alert in force shows red on the map (amber for a test) and on the megaphone, and is in the public alert feed. **Update** sends a new version to the same people that replaces it; **Cancel** sends the all-clear. Every alert, approval and delivery is in the ledger and on the incident's timeline.

> Sending to IPAWS itself (Wireless Emergency Alerts and the EAS) needs FEMA's agreement and the agency's own alert origination software; bitdispatcher writes its messages to the IPAWS profile for that software, and checks them with Google's CAP library and its IPAWS profile and with EAS Station's feed reader. No message has been sent to IPAWS.

CHAPTER 82

Public alerts: settings and lists

In the web interface: `/admin/publicalerts`

How public alerts are written and where they go (Admin › Public alerts). Administrators change the settings; supervisors keep the lists.

- **Sender**: the agency's CAP sender ID, an address it owns (alerts@county.gov), and its name.

- **Jurisdiction** and **county codes**: the six-digit SAME codes of the counties covered (0, the state's FIPS code and the county's: 051770 is Roanoke city), used for an alert to everyone and added to every alert for radio and TV.
- **Time zone**: CAP times carry the agency's offset.
- **Two-person approval** (recommended): a second supervisor approves each alert before it goes.
- **Public alert feed**: an Atom feed linking to each CAP message, and an IPAWS-style feed with the messages inside, for sirens, broadcasters' EAS encoders and apps. They show only alerts in force.
- **CAP receivers**: systems each alert is POSTed to as CAP (with a Bearer token); **Send a test** sends a test message.
- **Subscriber lists**: people who asked for alerts, reached through the paging gateways (text, email, push). A person with a place given gets only alerts whose area contains it. **Import a CSV** with the columns name, phone, email, topic, lang, lat, lon. A push gateway with no one listed publishes to its own topic, a channel anyone can follow.

> The feeds are public by design: they hold only what was sent to the public.

PART 6

Setting up

Everything is set up on these screens; there is no configuration file to edit.

CHAPTER 83

Self-service setup

An agency can set bitdispatcher up by itself, with no one from outside on site. Everything is configured on the admin screens, and the setup checklist (chapter 84) shows what is left. Support, whether from community support or a support agreement, is remote: e-mail, phone or a video call.

Who does what

- **Your administrator** works through the checklist on the admin screens: the map, stations, units, call types, users and connections.
- **Your IT staff** run the server (or two, for a second site (chapter 101)), its backups and its certificate.
- **Your phone, radio and paging vendors** connect their systems to bitdispatcher's documented interfaces: ALI spills or EIDO for calls, the station alerting gateway, paging gateways. Anything that needs hands, cables or a site visit is theirs.
- **Whoever supports you** answers questions and reads the support bundle (chapter 86) you send. No one needs to see your screen or sign in to your system.

The order that works

1. **Install** one server and sign in as the administrator. See Getting started (chapter 1).
2. **Work the checklist** from the top: Admin › Setup checklist. Each step opens its page and its chapter here. Open the checklist (chapter 84)
3. **Bring in the old system's data**: premises and hazards, call types and run cards and, if you want them, past incidents (Migration (chapter 103)); alarm permits (Alarm companies & permits (chapter 77)).

4. **Train** in Training mode: a copy of the system where nothing reaches real units. See Training (chapter 59).
5. **Run beside the old system** for a few weeks: enter calls in both, and compare the recommendations and the times.
6. **Pass the go-live check**: Admin › Go-live check. Fix what fails and run it again. Open the check (chapter 85)
7. **Switch over** at a quiet hour, with the old system still running read-only for a day.

Backups

The ledger (chapter 43) is the system: every setting, unit and incident is rebuilt from it. Copy the whole data directory off the server every night, keep several nights, and once in a while restore one on a spare computer and sign in. A second site is not a backup: a mistake is copied to both at once. The go-live check proves the ledger replays; it cannot see copies kept elsewhere, so the checklist asks you to confirm them.

> bitdispatcher sends nothing to BioDream or anyone else by itself. A support bundle leaves your system only when an administrator downloads it and chooses to send it.

CHAPTER 84

Setup checklist

In the web interface: `/admin/setup`

Admin › Setup checklist lists everything to do before the first live call, in order. bitdispatcher checks most steps itself as you work: a step is done when its page has what it needs.

- **Required** steps must be done before going live: the agency's name, the recorded map, stations, units, call types with run cards, the people (administrators with an authenticator), no demonstration data, nightly backups, training, and a passed go-live check (chapter 85) from the last week.
- **Recommended** steps make the system much more useful: the county's GIS, the phone system, station alerting and paging, a second site, running beside the old system, and who supports you.
- **Optional**: NCIC through the state message switch, which the state's CJIS office has to approve.

Each step says why it matters and what bitdispatcher found (*3 stations; not on the map: Station 4*). **Open** goes to the page that does it; the **?** opens its chapter.

Mark done is for what only a person can know: training given, backups copied off the server, the parallel run finished. Add a note if you like (*all three shifts*). **Not needed** is for a recommended or optional step the agency does without, with the reason (*no paging: stations are alerted by radio*). Steps bitdispatcher checks itself cannot be marked done by hand. Every mark is in the ledger with who made it; **Undo** takes it back.

CHAPTER 85

Go-live check

In the web interface: `/admin/golive`

Admin › Go-live check runs automatic tests before the first live call. Run it again after any big change: a new map, new stations, a new phone system. **Run the check** takes a few seconds; the result is in the ledger, and the setup checklist (chapter 84) counts it as done when nothing failed in the last week.

Each test passes, has something to look at, or fails:

- **The ledger is intact**: every event's hash and link, from the first.
- **The ledger restores**: a copy of the ledger is replayed on its own and compared with the running system (stations, units, call types, incidents).
- **Room on the disk**: fails under 2 GB free, warns under 10 GB.
- **No demonstration data**: no demo stations, premises, accounts or NCIC simulator.
- **Administrators use an authenticator** (CJIS asks for a second factor).
- **FIPS 140-3 cryptography**: passes when the server was built with `make fips` (the Go Cryptographic Module under FIPS 140-3 certificate #5247); warns otherwise. CJIS asks for validated cryptography for criminal justice information.
- **A trusted certificate**: warns when the server uses its own self-signed one.
- **The recorded map covers every station.**
- **Test calls at every station**: every call type with a run card, at every station that has units. Each is worked out and thrown away: nothing is written to the ledger and no unit is touched. It warns when drive times are straight lines, when another station's unit comes before the station's own (a response plan (chapter 93) can be the reason), and when a run card asks for units the agency does not have available.
- **The phone system is connected**, **station alerting and paging work** (a page delivered in the last 30 days) and **a second site is copying** (in the last 2 minutes).

Fix opens the page that fixes a test. The check never pages, texts or queries anyone. It cannot see backups kept elsewhere, the phone system's own wiring or the radio: confirm those on the checklist.

CHAPTER 86

Support and the support bundle

In the web interface: `/admin/support`

Admin › Support says who your agency asks for help and builds a support bundle to send them.

Who to ask for help is the agency's own text: the IT help desk, a support agreement's number and hours. Empty, it points to community support on GitHub (never post caller data, addresses or secrets there). Support for bitdispatcher is remote; anything that needs hands is done by your own staff and vendors.

Download the bundle builds one zip file with what a helper needs to find a problem without seeing your screen: the version, operating system and uptime; the ledger's size and check; disk space; the setup checklist and the last go-live check; the link to the phone system and the other sites; counts of units, stations and incidents and the kinds of events in the last 7 days; the settings, stations, units, call types and response plans; the connections; and the server's recent warnings and errors.

What is left out: calls, incidents, transcripts, recordings, caller names, numbers and locations, premise notes, usernames, passwords, tokens and keys. Every token, password and key in the connections becomes *(set)*. E-mail addresses, phone numbers and anything that looks like a token are removed from every file.

Read it before you send it. A name someone typed into a setting or a call type's notify list stays in. bitdispatcher sends nothing by itself: you download the bundle and decide whether to send it, and to whom. Each download is recorded in the ledger with who made it.

CHAPTER 87

Agencies

In the web interface: `/admin/agencies`

Every organisation the centre dispatches for is an **agency** (RPD, RCPD, SPD, RFD, RCFR, SFD, a public works department, and the centre itself): police, fire and EMS are disciplines, not agencies. A centre with no agencies works as one, with the centre-wide settings.

- **An agency**: its code and name, disciplines, active or not, ORI, NERIS ID and other identifiers, its own incident-number prefix, dispositions, status-check timers, talkgroups and records rules.

- **What belongs to it**: units, stations, call types and response plans name their agency (chosen from the list, not typed). An agency's response plan is only for its incidents; its timers only for its units.
- **People**: the agencies each person may work, the one they usually work for, those covered and watched by default, whether they are a **trainee**, and their roster entry. A person limited to some agencies changes only those agencies' configuration and sees only their supervisory views.
- **Workstations** can say which agencies they cover and watch (Admin › Workstations).

Change control

Configuration changes can be drafted on the training board, tried there, and **published** to the live board with a reason, all at once or **for one agency** only: a publish for one agency never carries another's or the centre's changes. Each publish keeps the versions it replaced: **Roll back** puts the earlier approved version back, with a reason, and is itself recorded.

CHAPTER 88

Operations rules

In the web interface: `/admin/operations`

The agency's rules for daily work (Admin › Operations rules; administrators).

- **Supervisory views**: the supervisor board, the operations report, quality review and care, the ledger and staffing coverage belong to the agency. Without a rule, supervisors see them; with one, only the roles, people and agencies it names. Administrators always see them. A person without the right neither sees the view nor reaches it by a direct link: the server refuses, and records the attempt.
- **Presets**: the agency's assigned preset for each duty (what trainees work and others start from), and protecting a person's preset so only administrators change it.
- **Working calls**: on or off, how many by duty and for trainees, and after how long a working call left by someone signed out is offered to supervisors.
- **Communications**: the kinds (who may send them, their default priority and expiry, whether they wait for people signed off), each priority's tone, repeat, acknowledgement and escalation (with a button to hear each tone), and the attachments allowed.
- **Staffing**: what happens to a unit left short (warn, limited, out of service), whether short units may be sent (warn, a supervisor's override, block), and the minimum qualified people the centre wants on duty.
- **Command line and keys**: the delimiter, the order of the identifiers for field initiate and unit status, the stop's call type, one or two passes, the automatic plate inquiry, several

units at once, the status codes (and retired ones), and the **key profile** for each duty, standard or classic (PremierOne-style) or the agency's own copy. **Check** finds a key doing two things on one screen, a key the browser keeps for itself, or an unknown action; the settings are refused until it is fixed.

CHAPTER 89

Units

In the web interface: `/admin/units`

Each unit has a call sign (`E1`, `M12`, `2A31`), a type (which gives its capabilities), a station, and optionally extra capabilities, the radio's unit ID and alias (so radio emergencies and positions find it) and an agency for mutual aid. New units start out of service.

CHAPTER 90

Stations

In the web interface: `/admin/stations`

Stations are where units are based. Their positions place units with no AVL, are where move-ups post covering units, and are where drive times start from when a unit's AVL is stale. Type the address and pick it from the recorded map's suggestions: the position fills in. Or type the position.

CHAPTER 91

Receiving facilities

In the web interface: `/admin/facilities`

Receiving facilities are the hospitals and specialty centres a patient may be taken to, and whether each is taking patients. Crews and the console see the status beside a transport decision.

- **Add** a facility with an id, name and kind (hospital, trauma, burn, stroke, cardiac, psychiatric or other), its phone and position.
- **Status** is a coloured chip: open (taking patients), on diversion, or closed. A quick button changes it in one click, and the change is an attributed ledger event.

Nothing here decides a destination; it tells the people who do what is open.

CHAPTER 92

Call types and run cards

In the web interface: `/admin/calltypes`

A call type is a kind of incident (`SF` structure fire, `CHEST` chest pain) with a default priority, a discipline and a run card: how many units of each capability it needs ("3 × engine, 1 × ladder, 1 × als"). Recommendations fill the run card. Capabilities come from unit types (Admin › Settings) and units.

Critical call puts the people who work one on the supervisor's care list (chapter 57); **Send a drone** has the drone launch rule ask for a drone (chapter 54).

Auto-dispatch (priority 3 to 5 types only): when no dispatcher has sent units within the wait in Settings (60 seconds unless changed), the recommended units go by themselves, logged as *auto-dispatch*. Only a complete recommendation without warnings is sent (a verified location, every line filled); otherwise the timeline says why and a dispatcher sends it. It tries once per incident.

CHAPTER 93

Response plans

In the web interface: `/admin/plans`

A **response plan** is a run order: the units an agency wants, in due order, when a plan applies. The recommendation follows a plan's due order when one applies; if a planned unit cannot go it says why and who goes instead. A person always decides.

A plan applies when all of its conditions match the incident:

- **Call types**: the selected codes (empty means every type).
- **Modifiers**: the selected modifiers, if any.
- **First-due station area**: a station's area, or Anywhere.
- **Addresses**: the incident address starts with one of these lines (one per line), for an address-specific plan.
- **From / To**: a time of day window (HH:MM), optional.

The **due order** is the unit ids in the order they are wanted; add from the list, reorder with the up and down buttons, remove any. A plan without an order narrows nothing and is ignored.

CHAPTER 94

Recording the map

In the web interface: `/admin/map`

bitdispatcher records the OpenStreetMap data for a circle around your centre into its own store, then draws the dispatch map, routes units by road and finds addresses from that store with no internet.

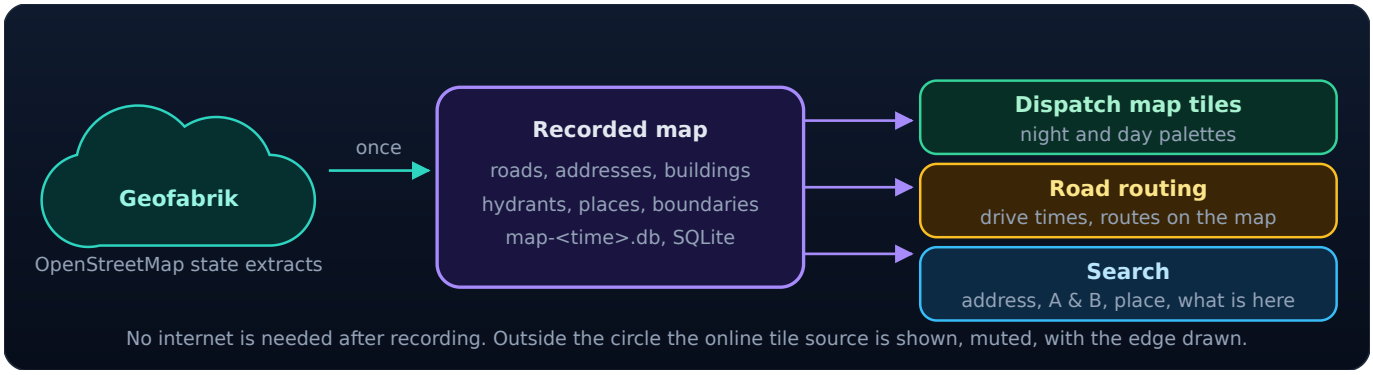


Figure 5. How the recorded map is made and used

1. Choose the centre (a preset, the console map centre, or a position) and the radius (100 miles covers a county and its mutual-aid neighbours).
2. The downloads the area needs are listed at once (state extracts from Geofabrik). Press **Record**.
3. Recording takes a few minutes; dispatching carries on with the current map, which stays until the new one is complete. Each recording is a ledger event.

Bring up to date applies OpenStreetMap's daily change files (Geofabrik publishes one a day per extract) since the map was recorded: a few megabytes instead of the whole extracts, then the store is built again. Record again for a new area, or when the map is more than a year behind; downloads already made are checked against their published checksums and reused.

Aerial photos

Below the map, **Aerial photos** records photos around the same centre: full detail within the radius you choose (zoom 19: rooftops and cars) and less out to the edge (zoom 16). The estimate (tiles, size, time) follows the form; tiles already recorded are skipped, so a stopped

recording carries on where it was. The sources and their terms are listed: VGIN's Virginia Base Mapping Program for Virginia, USGS The National Map (NAIP, public domain) elsewhere.

CHAPTER 95

County GIS

In the web interface: `/admin/gis`

The county's own GIS is more exact than OpenStreetMap where the county keeps it. Admin › County GIS imports its layers for the recorded map's area into a store of their own, kept when the map is recorded again.

Kind	Used for
Address points	Searched first: an exact county address point wins over OpenStreetMap's; house numbers on the map
Hydrants	Drawn with their IDs at street level
Emergency service zones, beats	Drawn and labelled on the map
Common place names	Searched
Road centrelines	Their names are searched; with address ranges (from/to, left/right) they place a number that has no point of its own, on its side of the road, and give a point on a road its number. Shown as **county range: estimate**
Overture addresses	Open address points from the Overture Maps Foundation (in the US largely the National Address Database), for any area, without a county service. Used after the county's own, marked **Overture**

1. Paste the layer's ArcGIS REST address (`…/FeatureServer/0`), choose a preset (Virginia's statewide address points and road centrelines from VGIN, or Overture's open addresses), or upload a GeoJSON file or a CSV with latitude and longitude columns. The layer is read at once: its name, how many records lie in the area, and a guess at the kind and the fields.
2. Check the field mapping against the sample values shown beside each field, and press **Add layer**.
3. **Import now** reads every layer that is on. The store in use stays until the new one is complete; each import is a ledger event.

Search results from county data are marked as such. ArcGIS layers are read with https only, in batches, so services that cap a query still give everything.

CHAPTER 96

Settings

In the web interface: `/admin/settings`

- **Agency** name and incident number prefix.
- **Status timers**: minutes in each status before the dispatcher is prompted for a status check.
- **Mark-outs**: the kinds a unit can be marked out for (code, name, status-check minutes, which units, whether it can be pulled for a call, whether it is retired, and the longest it may last before supervisors are told). See Police status and mark-outs.
- **Stale** ages for AVL and caller locations.
- **Shift changes** (`07:00 19:00`) for the operations report's shift periods.
- **Travel estimates** outside the recorded map: road factor and average speed.
- **Calls**: how near and how soon callers are grouped (chapter 50); the non-emergency (chapter 51) lines, words and reply.
- **People**: the critical-call (chapter 57) words, the support line, and when to suggest a break (chapter 58).
- **Quality review** standards and the questions to listen for (Quality review (chapter 61)); **crash data** thresholds (Vehicle crash data (chapter 53)); the line-attack (chapter 63) warning level.
- **Map beyond the recorded area**: the online tile source (fetched and cached by the server, never by browsers), and the map's starting centre and zoom.
- **Unit types**: code, name, discipline, capabilities, and whether stations keep them covered (move-up suggestions).
- **Dispositions** an incident can be closed with.

Sign-in notice is the system use notice shown on the sign-in page before anyone signs in: authorised use only, use is monitored and recorded (the CJIS Security Policy asks for one). **Auto-dispatch waits** is how long a call of an auto-dispatch type waits for a dispatcher (see Call types).

CHAPTER 97

Integrations

In the web interface: `/admin/integrations`

How bitdispatcher connects to call handling and radio. Each integration shows its live state and a checklist of the hooks in use.

- **Attachments** (under Connections): a **ClamAV daemon** (clamd, `host:port` or its socket) scans incident files and message attachments before they are kept; a file it reports is refused and recorded, and if clamd cannot be reached the upload is refused rather than kept unscanned. **Test** checks a clean file passes and the standard anti-virus test file is caught. **Read the words in pictures** uses the Tesseract program on the server for search (pictures never leave it); plain text and PDFs are read without it.
- **BitVoice**: its address and API token, and its certificate pinned with Read it. See BitVoice (chapter 98).
- **More BitVoice systems**: Add a BitVoice follows another one too, such as a neighbouring PSAP's: its name, address, API token, and its own EIDO token and radio secret (New makes them). Its calls show its name in the call queue, and what you do with one of them (text, rebid, video, photos) goes back to that BitVoice.
- **Neighbouring agencies**: their CADs, for sending and receiving incidents. See Incidents with neighbouring agencies (chapter 33).
- **Supplemental location (RapidSOS, NG9-1-1 LIS)**: a location server asked by HELD for where the caller's phone is: its address and the credentials the provider gives you, **Ask for every new 911 call**, and **Try a number**. The call-taker can press **Ask <name>** on the ANI / ALI panel. Not yet tested against RapidSOS itself (docs/SUPPLEMENTAL-LOCATION.md).
- **Public incident feed**: open incidents for the public or apps like PulsePoint: hundred-block addresses, points to about 100 m, no callers or notes; optional token and push address; **See what it shows** before turning it on.
- **EIDO**: the token a call-handling system uses to push EIDO to `/api/ng911/eido`.
- **Radio**: the secret for radio webhooks to `/api/radio/events`.
- **ALI spill**: the TCP address bitdispatcher listens on, and Test.
- **Hooks, drones and translation**: the hook token for transcripts, crash data, drone reports and radio words, the drone system, and the optional language model. See Hooks and connections (chapter 64).

Views (the place from every side, see Views (chapter 27)):

- **EagleView**: an application on EagleView's developer portal with Imagery API access under the agency's licence; its client ID and secret. **Sandbox** uses EagleView's test area (Omaha) to try it.

- **Panoramax**: no key; empty uses the catalogue of every instance, or give the agency's own instance.
- **Mapillary**: a client token from Mapillary's developer dashboard.
- **Google Street View link**: a link only.
- **Test at the map's centre** looks the centre up with each provider that is on and says what came back. Save first.

NERIS (fire and EMS reporting): the department's NERIS ID (FD and eight digits), a CAD integration's client ID and secret from NERIS, and the dispatch centre's name. With *Send each fire or EMS incident when it closes*, every closed incident with a fire or EMS call type or unit goes to NERIS as a CAD record (call, unit and clear times, the place, the disposition, the notes; police units left out); NERIS keeps it pending until the fire department completes it. **Check the last closed incident** asks NERIS to validate one without keeping it. Supervisors see *NERIS sent* (or why it was refused) on a closed incident, with **Send to NERIS** to send it again; a second send updates the same record. Use NERIS's test environment until NERIS has checked the integration.

CHAPTER 98

BitVoice

bitdispatcher uses every hook BitVoice offers a CAD, and needs none of them: without BitVoice, ALI and EIDO from any call-handling system work the same way.

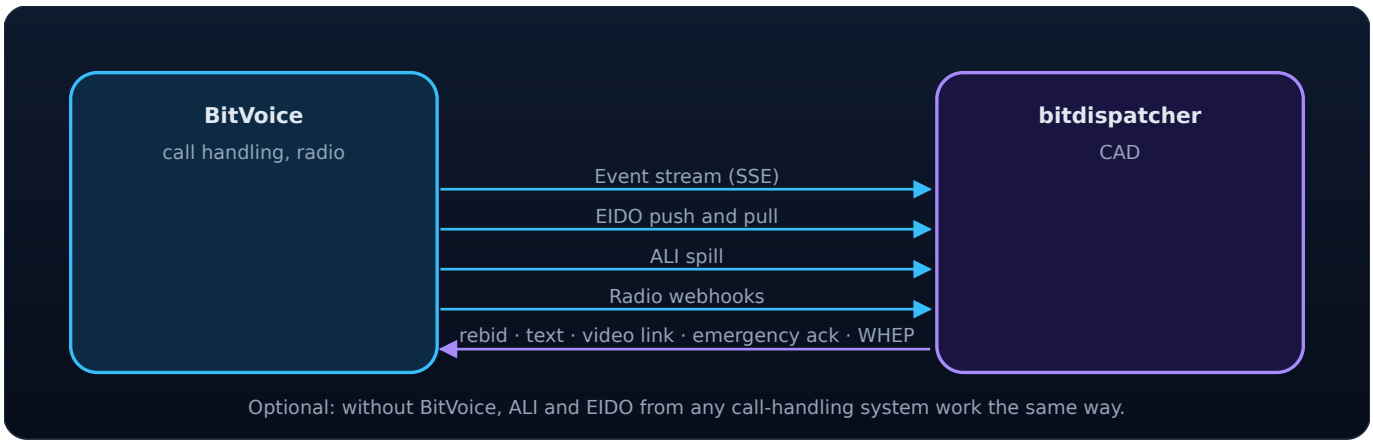


Figure 6. Every hook BitVoice offers a CAD, and which way each runs

Hook	Way	Used for
Event stream	BitVoice → bitdispatcher	Calls ringing, answered, ended; locations; resync after a gap
EIDO	both	Incident data pushed on answer and media changes; pulled when missed
ALI spill	BitVoice → bitdispatcher	The classic location record, merged with the same call
Radio webhooks	BitVoice → bitdispatcher	Emergency button, acknowledgement, transmissions, GPS
Callbacks	bitdispatcher → BitVoice	Rebid, text to caller, video link, live video, emergency acknowledgement

Delivery failures BitVoice reports show on the Situation panel.

What a 911 call from BitVoice shows on the call (incident panel and the call-taker's ANI / ALI):

- **number verified** (or **NOT verified**) from the caller ID's STIR/SHAKEN check, with the attestation. A failed check is a reason to doubt the number, not the emergency.
- **ESInet**, the service asked for, a **CAMA** legacy trunk and how its location was looked up, and the ways text came (MSRP, SIP, RTT, TTY).
- **Address in the address data** or **NOT**, from BitVoice's LVF: missing parts, *Did they mean…* corrections, and whether the caller's position agrees with the address.
- **Location check (BitVoice)** after a rebid: which location to show first and why.
- **Caller photos and clips**: each with its SHA-256, the camera's time and GPS, shown as facts until you press **Reveal (recorded)**; every reveal is in the ledger with who looked. **Hold as evidence** keeps them past BitVoice's retention.
- Caller video opens blurred and muted; **Reveal** is recorded in BitVoice's exposure record.
- A text-to-911 call waiting for a call-taker: **Take this text**, then reply in the box.
- When a call-taker passes the call on: BitVoice's briefing for whoever takes it, and who passed it on.
- After the call: the recording's name and its summary. The briefing and the summary are both kept, each marked as written by a language model.

If BitVoice ends a call because **the caller's audio stopped** (the line went silent and no hang-up came), the call says so and the advisory panel raises it as a silent 911 call: call back.

Reports › Operations report shows BitVoice's answer times (the NENA standard: 90% within 15 seconds), who looked at callers' photos and video, and the i3 log events received. The BitVoice light in the status bar turns amber when a PSAP the ESInet routes to is full, down or unstaffed.

CHAPTER 99

Users and positions

In the web interface: `/admin/users`

Each person holds one or more **duties**: call-taker, police, fire or EMS dispatcher, dispatcher (all disciplines), supervisor, administrator, records, field crew or viewer. They choose one at sign-in (chapter 2); the rights follow it. Their role, the most they may ever do, is the least that holds all their duties:

Role	Can
Viewer	See everything, change nothing (records staff are viewers who also work records)
Field crew	The mobile terminal (chapter 37) for its own unit
Dispatcher	Work incidents, units and calls
Supervisor	Also reopen incidents, read the ledger and reports, certify reports
Administrator	Also change configuration and users

One-time codes: the list shows who has an authenticator set up; **Reset** for a lost phone. You cannot change your own duties.

Passwords are generated or at least 12 characters. Five wrong passwords lock an account for 15 minutes. The positions list shows who is signed in where.

A public demonstration account. Tick **Public demonstration account** on a viewer, dispatcher or supervisor and the sign-in page offers **Sign in as the demo user**, with a choice of its duties: one press signs a visitor in, with no password, at a position of their own (WEB- and four letters) unless they typed one. One account at most, never an administrator or a field crew, and wrong passwords typed for it do not lock the button out. Use it only on a demonstration server: anyone who can reach the server can then work as that user.

CHAPTER 100

Workstations

In the web interface: `/admin/workstations`

A **workstation** is one of the agency's console positions. A browser enrolled as a workstation signs in there, and nobody is asked which console they are at: every action is recorded with the position already known.

- **Add** a workstation with a short id (`CT1`, `DISP2`) and, if you like, a name for where it sits.
- **Duties offered** narrow what a sign-in at that position may take: Call-taker, Police, Fire or EMS dispatcher, Dispatcher (all disciplines), Supervisor, Administrator, Records. Pick none to offer every duty.
- **Enrol this browser** makes the computer you are on that workstation; it is kept here, so it is remembered next time.
- **Get a code for another computer** shows a one-time code. On the other computer, open the sign-in page, choose "Enrol this computer" and type it. It works once, for 24 hours.
- **Enrolled browsers** are listed with their label and when; revoke any that should no longer sign in there.

Every enrolment and revocation is a ledger event.

CHAPTER 101

Several sites

In the web interface: `/admin/sites`

bitdispatcher runs at two or more sites, all live at once: a main centre and a backup centre, or neighbouring counties sharing a system. There is no primary to fail over from.

- **Each site writes its own ledger** and copies the others' over https, every second, each event checked against its site's hash chain before it is kept.
- **The same picture everywhere**: every site applies all events in the same order (by time, then site), so sites that have the same events show the same state, whatever order they arrived in.
- **Cut off, a site keeps working** on its own; when the link returns the events are exchanged and both sites settle on the same state. If two sites changed the same unit meanwhile, the later change wins, at both.
- **Incident numbers** carry each site's prefix (N26-000001, S26-000001), so two sites never give the same number.
- **Setting up**: give this site a name and prefix and save; copy its token and certificate fingerprint to the other sites. Add each other site with its address, its token and its

fingerprint (**Read it** reads the certificate; check it against the one shown on that site). The status bar shows how many sites are linked.

> A site's name and prefix are written into every event it makes, so they cannot change once in use.

CHAPTER 102

Premises and hazards

In the web interface: `/admin/premises`

A premise is what the centre knows about a place before anyone calls: **hazards** to the crews (a dog, oxygen in use, an armed occupant, a Knox box), **notes** and **contacts**. It is shown with every incident within 40 m of it, on the dispatcher's incident panel and on the mobile terminal, hazards first in red. An expired premise (a temporary hazard past its date) is shown as expired.

- **Add a premise**: type the address and pick it from the suggestions (that places it), click the common hazards or type others, add notes and contacts.
- **Search** by address, name, hazard or id; click one to change or remove it.
- Supervisors and administrators change premises; everyone sees them. Each change is a ledger event.
- **Pre-plan photos and views**: on a premise, add photos (with a caption and the side they show it from) and see the views (chapter 27) of the place. A photo's SHA-256 is in the ledger: a file changed on the server is refused, not shown.

Premises from an earlier system come in through **Migration**.

CHAPTER 103

Migration from an earlier CAD

In the web interface: `/admin/migrate`

Admin › Migration brings an earlier CAD system's data over from the spreadsheets it exports, CSV or tab-separated (UTF-8 or Windows encoding).

1. **Drop the file** on the page. Its columns are matched to ours from their names, and every row is checked at once.
2. **Choose what it holds** if the guess is wrong: premises and hazards, incident history, or call types and run cards. Change any column in the table; the first row's value is shown

beside it.

3. **Import** the rows that passed. The rows that did not are listed with why (no position and an address the map does not know, a date it cannot read).

Kind	Needs	Notes
Premises and hazards	an address (or latitude and longitude)	hazards and contacts separated by ; — an address without a position is found on the map (county points, then address ranges, then the map's)
Incident history	number and date	counts as calls before at an address: the incident panel shows the last five within 40 m over ten years, and the advisory panel's repeat-address rule counts them
Call types and run cards	code and run card	"2 engine; ladder:1; als": capabilities and counts. A capability no unit type has is pointed out

Importing the same file again replaces what it brought (premises by their id, call types by their code); history is added again, so import it once. Address points go to **County GIS**, which reads CSV files with latitude and longitude as well.

CHAPTER 104

Approved guidance

In the web interface: `/admin/guidance`

Approved guidance is the agency's policies, call-taking guides, response guides and checklists. The call-taker's Resources tab shows the matching **passages** with citations; nothing is invented, only approved text is shown.

A document has an id, title, **kind** (Policy (mandatory), Guide, Prompts or Checklist), version, effective date, an optional expiry, what it supersedes, and the source where the full approved copy is. Narrow it to call types, modifiers or words, or mark it **always shown**. Only a document marked **Approved for use** is shown to call-takers.

The **text** is Markdown with `#` / `##` / `###` headings. Headings divide it into passages. Under a heading, a line like `Applies: CARD, CHEST; modifiers: WEAPONS; words: not breathing` narrows that passage.

- **Add the examples** loads clearly-titled example documents as drafts (not approved), to see how it works.

- **Try it** matches a call type, modifiers and some text and shows the passages a call-taker would see, with their title, heading, kind, version and why, or a note when nothing matches.

A document's status is a chip: approved or draft, and outdated once its expiry has passed.

CHAPTER 105

Call-taking protocols

In the web interface: `/admin/protocols`

Your agency's question sets, worked on the call-taker's Entry tab once the call is submitted (docs/PROTOCOLS.md).

- **New protocol** or **Start from the example** (marked *not for use*). Give it an ID, a name and a version, and choose the call types it is offered for.
- **Questions**: each answer leads to the next question or ends at a **determinant**. **Make first** chooses where it starts.
- **Determinants**: a code, what it means, the priority and call type it sets (or keeps), and the instructions for the caller, one per line.
- Tick **Approved for calls** when your medical director or chief has approved it; drafts are not offered. Saving checks that every answer goes somewhere and nothing loops.
- On a call: one question at a time, answers on buttons or keys 1–9; every answer is on the timeline, and the determinant shows on the dispatcher's incident.
- Protocol software you license (ProQA and others) can post its determinant to the hook instead; it shows the same way, marked with the system.

CHAPTER 106

This manual

In the web interface: `/help`

This manual is built into bitdispatcher, so it is there wherever it runs, with no internet. Every screen's Help link opens its chapter; the palette (`Ctrl K`) finds chapters by name. The same chapters are printed as docs/manual/index.html and a PDF by `make manual` .

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